

FORGE: REACTION-GUIDED GENERATIVE DESIGN OF IONIZABLE LIPIDS

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ABSTRACT

Ionizable lipids are central components of lipid nanoparticles for RNA delivery. Their discovery often relies on synthesizing and screening libraries assembled from predefined reactants, which restricts the precursor structures available for design. Whole-molecule generation can extend this search, but verifying assembly requires precursor assignments and reaction information. We introduce FORGE (Flow-matched, Open-ended, Route-resolved Generation and Exploration), which uses known reaction chemistry to design ionizable lipids without restricting generation to a fixed reactant catalogue. Each reaction program specifies precursor roles, reaction-core positions, transformation order and coarse molecular architecture. FORGE generates the complete molecule at atom-and-bond resolution without component identifiers, stored precursor graphs or fragment tokens. Reverse decomposition and exact forward replay verify the implied assembly. We evaluate one shared model across 22 assembly families, including multicomponent condensations, repeated additions, ring openings and acylation reactions. Across [independent training seeds], verified exact level-one (exact-L1) yield is [macro mean and seed SD]%, compared with [matched null yield]% for whole-molecule generation followed by post-hoc verification. Structural checks, reference-lipid distributions and component-level diversity distinguish reaction consistency from molecular quality. The model generates [distinct component-novel yield]% exact-L1 products with recovered precursor identities absent from the training catalogues. Recursive synthesis assessment resolves complete precursor-preparation and terminal-material dossiers for [complete dossier yield]% of requests. These results [conclusion supported by conditioning, quality and route comparisons].

1 INTRODUCTION

Lipid nanoparticles (LNPs) deliver siRNA, mRNA vaccines and genome-editing reagents (Adams et al., 2018; Polack et al., 2020; Baden et al., 2021; Gillmore et al., 2021), with ionizable lipids supporting nucleic-acid complexation, particle assembly and endosomal escape (Cullis & Hope, 2017; Hou et al., 2021). Discovery of these lipids commonly relies on combinatorial synthesis and screening, in which modular precursors are assembled into libraries and the resulting formulations are evaluated (Akinc et al., 2008; Love et al., 2010; Li et al., 2024; Xu et al., 2024). This approach uses established reactions to combine known building blocks, which limits the lipids it can generate.

Generative methods incorporate assembly chemistry at different stages of molecular design. SynNet, RGFN and SynFlowNet retain reaction routes by selecting reactants from a supplied inventory and applying predefined transformations (Gao et al., 2022; Koziarski et al., 2024; Cretu et al., 2025), a strategy also used by lipid-specific synthesis-DAG and tree-search methods (Ou et al., 2024; Zhao et al., 2024). SynCoGen couples precursor design to assembly by jointly generating building-block graphs, reaction assignments and atomic coordinates (Rekesh et al., 2026). Whole-molecule models such as DeFoG and GenMol generate structures without a prescribed precursor inventory (Qin et al., 2025; Lee et al., 2025), leaving assembly and synthesizability to separate assessment (Gao & Coley, 2020).

FORGE learns a conditional distribution over complete lipid graphs from reaction programs that specify roles, reaction-core positions, transformation order and coarse architecture. It receives no

precursor identifiers, stored component graphs or fragment tokens. Reverse decomposition and exact forward replay verify each generated assembly. A bounded recursive search then assesses whether the implied precursors have preparation routes supported by reaction and supplier evidence, returning a computational dossier or an explicit unresolved outcome.

Our contributions are:

1. **Reaction-program-conditioned whole-molecule generation.** We developed a conditional discrete flow that generates complete lipid graphs with precursor identities left open, coupled to reverse decomposition and exact forward replay for assembly verification.
2. **Matched evaluation across 22 assembly families.** We compare one shared model with matched whole-molecule generation followed by post-hoc verification, conditioning controls and component-based generators. Family-wise assembly, decomposition precision and coverage, and held-component recovery quantify the contribution of reaction-program conditioning beyond the training inventory.
3. **Structural quality and synthesis evidence.** We assess headgroup retention, ring structure, chemical alerts, lipid-reference realism and product and precursor diversity alongside exact assembly. Recursive route assessment records precursor-preparation evidence, terminal-material support and unresolved outcomes for the generated identities, separating L1 consistency from complete synthesis dossiers.

2 PRELIMINARIES AND PROBLEM FORMULATION

2.1 IONIZABLE LIPIDS AND MOLECULAR ARCHITECTURE

Ionizable lipids combine protonatable amine-containing headgroups, linkers and hydrophobic tails (Semple et al., 2010; Jayaraman et al., 2012). Protonation supports nucleic-acid association during acidic formulation and membrane interactions involved in endosomal release (Xu et al., 2024). LNPs commonly combine an ionizable lipid with helper lipids, cholesterol and PEGylated lipids. Lipid structure, stereochemistry and formulation composition affect delivery (Han et al., 2021; Hatit et al., 2023; Cheng et al., 2020); molecular constitution and formulated-particle performance are therefore distinct evaluation levels.

2.2 REACTION FAMILIES AND MODULAR ASSEMBLY

We study 22 registered assembly families (Table 21). These include multicomponent condensations, repeated amine additions, ring-opening reactions, esterification and amidation. Ugi 3-CR combines an amine, aldehyde and isocyanide into an α -amino amide (Xu et al., 2024). Repeated aza-Michael addition attaches acrylate-bearing components to amine sites (Li et al., 2023); repeated reductive amination couples aldehyde-bearing components to amines through condensation and reduction (Xue et al., 2024). The Ugi variant has no carboxylic-acid reactant component and uses an acidic phosphorus catalyst; its ester functionality comes from the aldehyde precursor. Repeated assemblies require both transformation order and participating sites. Precursor-origin boundaries can span functional headgroup, linker and tail regions. These distinctions motivate role-assigned reaction programs (Appendix D.1, Table 21).

2.3 MOLECULAR GRAPHS AND REACTION PROGRAMS

A lipid constitution is a nonempty, connected, simple molecular graph $G = (V, E, \ell_V, \ell_E)$, with atom labels $\ell_V : V \rightarrow \Sigma_{\text{atom}}$ and bond labels $\ell_E : E \rightarrow \Sigma_{\text{bond}}$ in finite vocabularies. Bond order is an edge label. We compare graphs up to label-preserving isomorphism after removing atom-map and stereochemical labels, retaining formal charges and bond orders. We write G for a graph or its constitutional isomorphism class when the distinction is immaterial. The bounded graph space $\mathfrak{G}_{N, K_{\text{cl}}}$ contains these classes with $1 \leq |V| \leq N$ and cycle rank $\beta(G) = |E| - |V| + 1 \leq K_{\text{cl}}$. This is a structural support definition; molecular validity additionally requires valence and sanitization checks. Here $N = N_{\text{max}} = 254$ and $K_{\text{cl}} = 12$.

A reaction program specifies the assembly constraints,

$$P = (c, d, \mathbf{r}, \mathbf{k}, M). \quad (1)$$

Here c identifies a registered assembly chemistry with its admitted transformation sequences, d is the requested depth, r assigns precursor roles to atom positions, and k marks reaction-core positions. The four morphology fields in M record each role’s exterior atom count, junction budget, closure count and core-attachment count. These quantities constrain coarse architecture without specifying precursor identities. Precursor origin and core membership are separate: an atom can belong to a precursor and participate in the core; a template-introduced atom belongs to the core without belonging to a precursor. Appendix A.1 defines the morphology counts.

2.4 EXACT ASSEMBLY, NOVELTY AND ROUTE EVIDENCE

For registered chemistry c , a precursor input $\mathbf{C} \in \mathcal{C}_c$ is an ordered tuple of constitutions, roles and transformation steps. Its registered forward products within the bounded support form $\mathcal{F}_c(\mathbf{C}) \subseteq \mathcal{G}_{N, K_{cl}}$. Reverse search returns a finite set $\mathcal{D}_c(G)$ of precursor inputs and assembly traces $(\mathbf{C}, \tau)$. Exact forward replay retains

$$\hat{\mathcal{E}}_c(G) = \{(\mathbf{C}, \tau) \in \mathcal{D}_c(G) : G \in \mathcal{F}_c(\mathbf{C})\}. \quad (2)$$

Membership uses constitutional identity. A valid graph passes verified exact level-one assembly (L1) when $\hat{\mathcal{E}}_c(G) \neq \emptyset$. Bounded reverse search can miss assemblies, and recovered traces need not match every sampled depth, role-position or morphology field in P . Assembly existence and conditioning adherence are distinct checks, as are decomposition coverage and replay precision (Appendix A.5).

A recovered precursor is component-novel when its constitution is absent from its role’s catalogue. Strong product-level catalogue escape requires *every* exact precursor input to contain an outside component. We measure recovered component novelty, which does not establish that universal claim (Appendix C.2). A complete computational synthesis dossier also requires evidence-qualified precursor preparation (L2) and terminal-material availability (L3). Incomplete assessments remain unresolved or search-censored; these checks do not establish experimental synthesis success or delivery activity.

3 FORGE

3.1 OVERVIEW

FORGE learns the distribution $p_\theta(G | P)$ of complete lipid molecules G given a reaction program P . The model receives no precursor identifiers, stored component graphs or fragment tokens. The program specifies how the molecule is assembled without selecting the precursor structures. FORGE has four main steps (Figure 1):

1. **Specify the reaction program.** Choose assembly chemistry, roles, reaction-core positions, transformation order and morphology, leaving precursor identities open.
2. **Generate the whole graph.** A conditional discrete flow generates atom, bond and topology states while preserving program-fixed coordinates.
3. **Verify exact assembly.** Recover role-assigned precursors and require their forward transform to reconstruct the generated molecule.
4. **Construct precursor routes.** Search for preparation routes to evidence-qualified starting materials and abstain when a complete dossier cannot be constructed.

3.2 CONDITIONING ON REACTION PROGRAMS

An admissible complete program is sampled from the training-fold prior,

$$p_\theta(G, P | c) = p_{\text{prog}}(P | c) p_\theta(G | P). \quad (3)$$

Learned embeddings combine chemistry, depth, precursor role, core status and role-local morphology at each graph position. The adapter determines active positions and fixed or variable coordinate masks. Appendix A.1 specifies the embeddings and morphology counts.

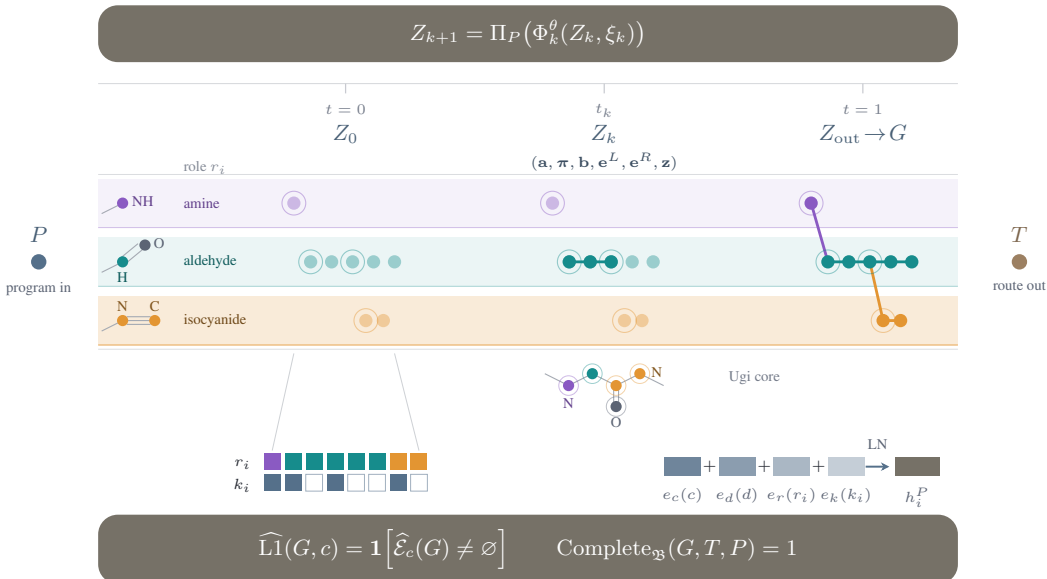


Figure 1: **FORGE generation and route assessment.** A reaction program fixes assembly roles and core coordinates without selecting precursor identities. FORGE generates the whole graph, verifies exact assembly replay, and recursively traces its implied precursors to evidence-qualified materials or an explicit abstention. Colours mark precursor origin; grey denotes the Ugi template-introduced oxygen. Replay and route closure are computational checks, not experimental synthesis evidence.

3.3 GRAPH REPRESENTATION AND DISCRETE FLOW

FORGE represents each molecule as a spanning tree with additional bonds for ring closures. A deterministic encoder maps the molecular graph G and its selected atom-mapped assembly annotation α to

$$X_{\star} := \text{Enc}_P(G, \alpha) = (\mathbf{a}, \boldsymbol{\pi}, \mathbf{b}, \mathbf{e}^L, \mathbf{e}^R, \mathbf{z}), \quad (4)$$

where $\mathbf{a}$ records atom states, $\boldsymbol{\pi}$ identifies each atom’s parent in the tree, and $\mathbf{b}$ records the tree bonds. The arrays $\mathbf{e}^L$ and $\mathbf{e}^R$ identify the endpoints of additional bonds, whose states are stored in $\mathbf{z}$.

The padded state space $\bar{\mathfrak{X}}_{N, K_{\text{cl}}}$ is finite. Its valid encodings form $\mathfrak{X}_{N, K_{\text{cl}}}$: each contains a connected tree and at most K_{cl} distinct closure edges. The decoder Dec reconstructs the graph from a valid encoding and returns $\perp_{\text{dec}}$ for a malformed state. Every graph within the specified atom, bond, size and closure limits has an encoding (Theorem C.3).

During sampling, we also restrict the allowed parents and number of children for each atom. These constraints limit which valid encodings the sampler can use.

For a given P , let $\mathcal{I}(P)$ index the categorical coordinates, each with finite alphabet $\mathcal{X}_i(P)$. Pointer alphabets contain only positions permitted by the layout. The adapter partitions coordinates into fixed positions $\mathcal{F}(P)$ with values $x_{P,i}$ and variable positions $\mathcal{V}(P)$. The set Ω_P contains raw states within the allowed alphabets that satisfy the fixed values. It can contain malformed or chemically invalid states. The source law q_0^P fixes $\mathcal{F}(P)$ and uses independent, full-support marginals $q_{0,i}^P$ on variable alphabets. For a clean endpoint $x_1 \in \Omega_P$, the conditional corruption law is

$$q_{t|1}^P(x \mid x_1) = \mathbf{1}[x_{\mathcal{F}(P)} = x_{P,\mathcal{F}(P)}] \prod_{i \in \mathcal{V}(P)} [(1-t)q_{0,i}^P(x_i) + t\mathbf{1}\{x_i = x_{1,i}\}], \quad 0 \leq t \leq 1. \quad (5)$$

Each coordinate interpolates between its source and clean value. Their conditional joint law is a product, so it generally differs from a linear mixture of the joint source and endpoint laws.

The clean-coordinate predictor $d_{\theta,i}(a \mid x, t, P)$ assigns a probability to each $a \in \mathcal{X}_i(P)$. A continuous-time Markov chain (CTMC) uses these predictions to average conditional one-coordinate

rates,

$$R_{t,i}^\theta(u, v \mid x, P) = \sum_{a \in \mathcal{X}_i(P)} d_{\theta,i}(a \mid x, t, P) R_{t,i}^*(u, v \mid a, P), \quad u \neq v, \quad t < 1. \quad (6)$$

The conditional rate $R_{t,i}^*$ realizes the corresponding factor in Equation 5; Appendix C.3 defines it and proves the path identity. The joint generator permits one coordinate to change at an infinitesimal time. Coordinate posterior marginals therefore suffice to specify its rates, even when endpoint coordinates are statistically dependent.

3.4 PROGRAM-CONDITIONED TRANSFORMER AND TRAINING

A graph Transformer predicts clean coordinates from (X_t, t, P) . Each layer combines graph self-attention with parent, child and closure relation biases and cross-attention to program, depth, role and core tokens. Softly program-routed residual adapters add family-specific parameters. Qualified atom-mapped traces supervise role and core annotations; unresolved source-trace ambiguity excludes a record from semantic training.

The source-balanced training law μ_{bal} covers annotated triples (G, α, P) with equal mass across the 22 qualified families and uniform sampling over unique constitutions within each family, with replacement. No second loss reweighting changes this measure. The model combines masked coordinate cross-entropy, role-balanced chemistry, role/core supervision, and repeat and topology consistency terms. We combined chemistry-task gradients with PCGrad (Yu et al., 2020). Appendix A.3 specifies the full objective and training-time distribution.

To isolate the statistical property of coordinate denoising, define the reference population loss

$$\mathcal{L}_{\text{ref}}(\theta) = \mathbb{E}_{(G, \alpha, P) \sim \mu_{\text{bal}}, t \sim \mathcal{U}(0,1)} \left[\sum_{\substack{X_t \sim q_{t|1}^P(\cdot | X_*) \\ i \in \mathcal{V}(P)}} w_i(P) [-\log d_{\theta,i}(X_{*,i} \mid X_t, t, P)] \right], \quad w_i(P) > 0. \quad (7)$$

The weights depend only on the program; they can normalize coordinate counts without depending on the unknown clean category. This reference loss supports the population analysis. It is not the complete production training objective.

The reference objective has an ideal population interpretation. For a program with positive training mass, let q_1^P be the law of $\text{Enc}_P(G, \alpha)$ under $\mu_{\text{bal}}(\cdot \mid P)$, and let μ_G^P be the corresponding graph law.

Theorem 3.1 (Ideal conditional generation). *Fix such a program P , with registered chemistry c , finite coordinate alphabets and a full-support source on each variable alphabet. Assume every training endpoint belongs to Ω_P , decodes to its source graph, and passes molecular validity and verified exact LI. Assume an unrestricted measurable coordinate predictor attains a population minimum of Equation 7. Starting from q_0^P , exactly simulate the local CTMC of Equation 6 on $0 \leq t < 1$. Then its marginal law converges to q_1^P as $t \uparrow 1$, and*

$$\text{Dec}_\# q_1^P = \mu_G^P, \quad \Pr_{G \sim \mu_G^P} [\text{Valid}(G) = 1, \widehat{\text{LI}}(G, c) = 1] = 1. \quad (8)$$

Here $\text{Dec}_\#$ denotes the distribution obtained by decoding.

Appendix C.3 proves this result by recovering the clean-coordinate posteriors and verifying the CTMC forward equation. The theorem concerns the reference population objective and the limit of exact continuous-time sampling. It establishes neither generalization beyond q_1^P nor correctness of finite training, PCGrad, capped Euler updates or constrained terminal readout.

3.5 SAMPLING AND EXACT ASSEMBLY VERIFICATION

The numerical sampler uses states Z_k , distinct from the ideal CTMC X_t . Starting from $Z_0 \sim q_0^P$, it applies

$$Z_{k+1} = \Pi_P(\Phi_k^\theta(Z_k, \xi_k)), \quad k = 0, \dots, L_{\text{samp}} - 1, \quad (9)$$

where Φ_k^θ is a capped independent-coordinate Euler update, ξ_k contains its random draws and Π_P restores program-fixed coordinates. Fixed coordinates have zero rate and remain constant (Proposition C.6); chemical consistency requires verification. A fresh prediction at $t = 1$ drives the constrained argmax readout $Z_{\text{out}} = H_P^\theta(Z_{L_{\text{samp}}})$, including family-specific core and reactive-site constraints, or a typed failure. The checked inference arm augments candidate graphs with qualified structural proposals and selects under explicit product and precursor diversity constraints. We reassess the final graph and retain raw outputs as a separate arm; Table 9 distinguishes these inference contributions from the learned generator. We then decode, sanitize and apply Equation 2 to each valid graph. Molecular validation, decomposition, exact replay and root ambiguity are recorded separately. Appendix A.4 specifies the numerical operations and their distinction from the ideal CTMC.

3.6 RECURSIVE ROUTE ASSESSMENT

For a molecule with exactly one verified precursor decomposition (C_T, τ_T) , we search for routes to prepare its precursors within fixed depth and query limits:

$$Y = \text{Assess}_{\mathfrak{B}}(G, P, C_T, \tau_T) \in \mathfrak{D} \cup \mathfrak{F}_{\mathfrak{B}}. \quad (10)$$

Here $\mathfrak{B}$ is the frozen route contract, $\mathfrak{D}$ the dossier space and $\mathfrak{F}_{\mathfrak{B}}$ its finite typed outcome set. Ambiguous roots cause abstention. A complete dossier requires molecular validity, exact L1, admitted and verified upstream reactions, and available terminal materials. Other outcomes are unresolved or search-censored. Assessment operates on the completed graph without modifying generation. Appendix A.6 and Algorithm 1 specify admission and inference; Appendix C states the formal guarantees and their assumptions.

4 RESULTS

4.1 EVALUATION PROTOCOL AND MATCHED BUDGETS

The evaluation compares correctly conditioned, shared-null and cyclically mislabelled models under matched architecture, training exposure, checkpoint selection and inference budgets. The evaluation population contains [held-out requests per family per seed] requests per family across [independent training seeds] independent training seeds. Component-disjoint held-out products are excluded from training, decoder development and checkpoint selection. Verified exact-L1 yield counts attempts passing molecular validation, decomposition and exact forward replay, with *all* attempts in the denominator. Post-hoc rejection cannot increase the proposal’s per-attempt acceptance probability (Proposition C.10). We report means and sample standard deviations across training seeds, distinguishing retraining variation from conditional Monte Carlo variation. Raw and final yield count one output per request before and after the declared inference procedure; failed searches and abstentions remain in the denominator. Appendix B specifies the splits, source domains, proposal counts and computational budgets.

4.2 SHARED GENERATION ACROSS 22 ASSEMBLY FAMILIES

We first ask whether one conditional flow can cover 22 assembly families while maintaining performance on the matched three-family comparison. The equal-family exact-L1 yield is [raw macro L1] for raw readout and [final macro L1] after checked inference (Table 1). The lowest family yield is [lowest family L1], with [between-family pattern]. Decomposition coverage, candidate precision and ambiguity are [decomposition outcomes]; replay of accepted traces is [accepted-trace replay]. These quantities distinguish an inability to recover a valid assembly from an inconsistent returned trace. Fixed-coordinate violations and support overflows occur in [coordinate and support failures] of [total requests] requests. On common requests for Ugi 3-CR, aza-Michael addition and reductive amination, the change relative to the three-family model is [three-family bridge effect] (Table 8).

Table 1: **Exact assembly across 22 ionizable-lipid reaction families.** Values are mean $\pm$ sample standard deviation across independent training seeds, with all requests in the yield denominator. Raw and final outputs share request identities; final includes checked completion, repair and selection. Null and cyclic arms use matched training and inference budgets. Decomposition coverage and candidate precision are distinct from replay of accepted traces. Per-seed counts, replay invariants and ambiguity: Appendix B.

Family	Raw L1 (%)	Final L1 (%)	Shared null (%)	Cyclic label (%)	Decomp. cov. / candidate prec.
A3 amine-aldehyde-alkyne	—	—	—	—	—
Acid-epoxide diester	—	—	—	—	—
AEMA	—	—	—	—	—
Aldehyde Ugi 3-CR	—	—	—	—	—
Aldehyde Ugi 4-CR	—	—	—	—	—
α -Isocyanoester	—	—	—	—	—
dihydroimidazole	—	—	—	—	—
Amine alkylation	—	—	—	—	—
Amine-epoxide	—	—	—	—	—
Aryl reductive amination	—	—	—	—	—
Aza-Michael acrylamide	—	—	—	—	—
Aza-Michael acrylate	—	—	—	—	—
Disulfide Michael	—	—	—	—	—
Epoxide O-acylation	—	—	—	—	—
iPhos	—	—	—	—	—
Ketone-isocyanide amide	—	—	—	—	—
Ketone Ugi 4-CR	—	—	—	—	—
Maleate	—	—	—	—	—
O-esterification	—	—	—	—	—
Passerini	—	—	—	—	—
Preassembled thiol-yne	—	—	—	—	—
Reductive amination	—	—	—	—	—
Thiolactone	—	—	—	—	—
Equal-family macro	—	—	—	—	—
Pooled all requests	—	—	—	—	—
Worst-family result	—	—	—	—	—

4.3 CONTRIBUTION OF REACTION-PROGRAM CONDITIONING

To isolate the effect of reaction information during generation, we compare the conditioned model with the shared-null and cyclic arms under the same attempt budget and post-hoc verifier. The conditioned-minus-null difference is [conditioning difference] percentage points, with [conditioning uncertainty and family variation]. The cyclic control retains precursor roles, reaction-core coordinates and program depth, isolating the contribution of the reaction-identity label within a structured program. Its difference from correct conditioning is [identity-label difference]. The information supplied through layouts, masks, cores and decoder priors is accounted for separately (Table 8); the resulting attribution is [program versus identity-label conclusion].

4.4 DECODER AND ARCHITECTURE EFFECTS ACROSS PROGRAMS

With model weights fixed, the raw-to-final exact-L1 change is [decoder L1 difference], and the change in outputs satisfying the qualified structural checks is [decoder design difference] (Table 9). The same source verifier assesses core-constrained readout, component completion, scoped head retention, role-local atom/bond proposals and final ring-system repair. Original candidates remain available to selection. Reselection within the original pool changes yield by [selection-only difference]; adding repaired candidates contributes [repair-pool difference], at a cost of [additional proposal and source-check cost]. Larger-pool gains and equal-cost contrasts are reported sepa-

rately, because additional search and a changed selection objective do not isolate the effect of a repair rule.

The architecture comparison separates joint denoising from parameter-matched independent precursor roles (FACT), cross-attention and gradient-conflict control. Exact-L1 and structural-quality contrasts are [architecture contrasts], with [architecture family dependence] (Table 18). These comparisons separate learned-model effects from changes made at inference; motif agreement with the same TRAIN components that define a decoder prior is not independent validation.

4.5 PRECURSOR NOVELTY, DIVERSITY AND HELD-COMPONENT GENERALIZATION

We next ask whether FORGE can preserve exact assembly while generating precursor constitutions absent from training. Verified component-novel yield is [component-novel yield] per 1,000 requests, including [distinct component-novel yield] distinct products (Table 15). Finite-inventory methods have zero component novelty relative to their own visible inventory by construction. Component novelty is verifier-relative and does not establish product-level catalogue escape. Exact product reachability gives [catalogue reachability outcome]. Role-specific unique counts and Shannon and Simpson effective counts are [component diversity outcome], distinguishing a broad precursor population from repeated use of a few components.

Held-component recovery is [held-component yield] per 1,000 requests, covering [held-identity coverage] of reserved identities. Source-disjoint and component-disjoint exact yields are [generalization yields], with [size and repeat effects] across molecular size, closure count and repeated-arm strata (Table 16). Comparisons with external generators use their qualified chemical scope and a common verifier; their outcome is [external-baseline comparison]. Unsupported native families remain unassessed. Fresh noise applied to TRAIN-derived requests is a sampling diagnostic, not evidence of held-out generalization.

4.6 STRUCTURAL QUALITY AND AGREEMENT WITH LIPID REFERENCES

Exact assembly and lipid structural quality measure different properties. The fraction satisfying exact L1 and all applicable design checks is [raw joint design yield] for raw outputs and [final joint design yield] after inference (Table 2). Chemical alerts, loss of a required headgroup site and ring-condition violations account for [structural failure distribution]. These checks apply to the complete final graph; source-positive controls give [control disagreement and abstention rates]. Chemistry-context flags remain separate from confirmed defects. A retained-basic-site query does not establish pKa, and absence of a narrow alert does not establish physical stability or delivery activity.

Table 2: **Structural quality and agreement with design conditions.** Raw and final arms share requests; values are mean $\pm$ sample standard deviation across training seeds. Controls are source-positive structures within each metric’s qualified scope. Applicable and assessed counts, control disagreements and abstentions accompany conditional rates; unsupported checks are not counted as passes.

Metric	Raw	Final	Controls
Valid molecules / all requests (%)	—	—	—
Connected molecules / all requests (%)	—	—	—
Exact L1 + all qualified design checks / requests (%)	—	—	—
Molecules with a justified chemical alert / requests (%)	—	—	—
Molecules requiring chemistry-context review / requests (%)	—	—	—
Required head retained / family-scope requests (%)	—	—	—
Correct role-local cycle allocation / requests (%)	—	—	—
Requested ring-size agreement / comparable requests (%)	—	—	—
TRAIN atom-environment coverage (%)	—	—	—
Ring-system TRAIN support / assessed products (%)	—	—	—
Exact L1 + product/component features observed / requests (%)	—	—	—

TRAIN atom/ring support is a reference diagnostic and may share information with decoder priors. Definitions: Table 13; per-family outcomes: Tables 10 and 11.

Agreement with lipid references is [realism outcome] in fingerprint and descriptor spaces (Table 3). Precision measures proximity to the reference population, whereas coverage measures how much of that population is represented. Their joint pattern is [precision-coverage interpretation]. Nearest-reference similarity and internal diversity distinguish concentrated familiar outputs from diverse but distant structures. The source-study-grouped two-sample classifier has AUC [two-sample AUC], with [two-sample uncertainty]. These comparisons distinguish TRAIN support from independent lipid-reference agreement; near-chance discrimination does not prove chemical quality.

Table 3: **Lipid realism and distributional coverage.** Raw and final outputs are compared within each family and reference population, with unique-molecule and all-request denominators distinguished. TRAIN-reference diagnostics are separate from comparisons with source-disjoint observed lipids. Values are mean $\pm$ sample standard deviation across training seeds.

Metric	Raw	Final	Reference
Fingerprint precision (%)	—	—	Declared lipid cohort
Fingerprint coverage (%)	—	—	Declared lipid cohort
Fingerprint recall (%)	—	—	Declared lipid cohort
Distinct reference-supported products / requests (%)	—	—	Declared lipid cohort
Nearest-reference Tanimoto similarity	—	—	Declared lipid cohort
Internal ECFP4 diversity	—	—	Generated population
Descriptor distribution distance, per feature	—	—	Declared lipid cohort
Descriptor-space precision and coverage	—	—	Source-disjoint lipids
Source-grouped two-sample classifier AUC	—	—	Source-disjoint lipids

ECFP4 uses Morgan radius 2 and 2,048 bits; neighborhood metrics use five neighbors. Unsupported reference comparisons remain unassessed. Table 14 gives the full descriptor panel; Table 12 reports each family and reference population.

4.7 RECURSIVE SYNTHESIS EVIDENCE AND COMPLETE-ROUTE YIELD

Exact final assembly closes L1 but does not establish an upstream route. Exact precursor-preparation evidence (L2) covers [L2 component coverage] of required components, and qualified terminal evidence (L3) covers [current L3 coverage] (Table 17). Complete-product dossier yield is [complete dossier yield] per 1,000 requests, with [family-wise evidence depth]. Each complete dossier requires every precursor branch to terminate in an admitted exact material, preserving role, reaction-step and repeated-component context. Directly accepted terminals do not require an L2 synthesis. Missing or scope-mismatched sources, unresolved identities, expired availability and bounded-search censoring are reported separately. The dominant limitation is [route-closure limitation]; absence of admitted evidence is not evidence of unsynthesizability. These results concern auditable computational synthesis dossiers and do not establish experimental synthesis or biological activity.

5 DISCUSSION

FORGE incorporates synthesis constraints into whole-molecule generation without restricting the neural generator to a fixed precursor catalogue. Across 22 reaction families, program conditioning changes exact-assembly yield by [paired conditioned-minus-null contrast] percentage points. Joint whole-molecule generation and component-factorized generation differ by [family-wise joint-generation contrast], indicating [interpretation of coordination across precursor-derived regions].

Leaving precursor identities open permits verified assemblies to contain components outside the training catalogue. The observed component-novel yield of [yield]% and held-component recovery of [recovery]% distinguish exploration beyond observed identities from recovery of specifically withheld structures. Component novelty alone does not establish that a product lies outside every catalogue-reachable assembly.

Exact assembly is one level of evidence. Chemical alerts, retained head sites, ring conditions, reference-distribution coverage and component diversity assess different properties of the generated lipids. Their joint results show [quality and diversity conclusion], with residual failures concentrated in [families and structural failure modes]. TRAIN-derived structural priors contribute [inference

gain and cost]; reference support computed from those same priors cannot independently establish realism. Documentary L2/L3 assessment closes [complete dossier yield]% of requests and leaves [unresolved and search-censored proportions]. These computational checks do not establish experimental synthesis success, ionization or delivery performance.

AI USE STATEMENT

Generative AI tools assisted with conceptual and methodological development, software implementation and debugging, experimental design and diagnostic analysis, interpretation of computational results, literature search and summarization, and drafting and editing the manuscript. The authors are responsible for the final code, reported results, citations and conclusions, including material produced with AI assistance.

ETHICS STATEMENT

This work evaluates ionizable-lipid generation and synthesis evidence computationally. Molecular design systems can be misused. Chemical validity, exact assembly, synthesis-route evidence and predicted properties do not establish safety or biological efficacy. Generated structures require appropriate evaluation before experimental use.

REPRODUCIBILITY STATEMENT

The Methods and Appendix specify the molecular representation, conditional discrete flow, numerical sampler, exact assembly verification, route assessment, benchmark protocols and evaluation metrics. Appendix A.8 records the source, configuration, split, checkpoint and evaluation artifacts. Per-seed results appear in Table 7; assumptions and proofs appear in Appendix C.

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A METHOD AND IMPLEMENTATION DETAILS

This appendix specifies program and graph encoding, model training, sampling, verification and route assessment. Algorithm 1 combines the inference steps.

A.1 REACTION-PROGRAM ENCODING

The program tuple in Equation 1 uses the ordered atom-position set $\mathcal{I}_{\text{atom}}(P) = \{1, \dots, n(P)\}$. The arrays $\mathbf{r}$ and $\mathbf{k}$ assign precursor roles and core positions to these atoms. This atom-position index differs from $\mathcal{I}(P)$, which indexes all six families of categorical coordinates.

Morphology is computed from the selected atom-mapped trace and its sparse serialization. For role r , let V_r^{ext} contain its exterior atoms, excluding reaction-core atoms, and let $o_r(v)$ count the same-role exterior children of v in the serialized tree. The four fields are

$$M_r = (n_r, j_r, \gamma_r, a_r), \quad n_r = |V_r^{\text{ext}}|, \quad j_r = \sum_{v \in V_r^{\text{ext}}} \max\{o_r(v) - 1, 0\}. \quad (11)$$

Here γ_r counts closure edges with both endpoints in V_r^{ext} , and a_r counts same-role bonds between exterior and core atoms. When the serialized tree restricts to a spanning forest of the exterior subgraph, γ_r equals that subgraph’s cycle rank. The junction budget is a tree-branching count; it does not count degree contributed by closure bonds. Consequently these fields depend on the selected annotation and serialization, not on the unannotated graph alone. Fixed-coordinate equality alone does not establish agreement with every morphology field. Positions without precursor origin use the designated absent morphology state. Integer morphology values are encoded with an offset of one, reserving zero for that absent state.

The training-fold program prior is defined in Section 3.2. The memory encoder uses separate program-identity and depth tokens, $m_c = e_c(c) + e_{\text{type}}(0)$ and $m_d = e_d(d) + e_{\text{type}}(1)$, together with one token per active atom position,

$$m_i = e_r(r_i) + e_k(k_i) + e_{\text{pos}}(i) + e_{\text{rep}}(g_i) + e_{\text{comp}}(u_i) + \sum_{v=1}^4 e_{M,v}(M_{r_i,v}) + e_{\text{type}}(2). \quad (12)$$

Here g_i is the repeat-group state and u_i the position within the precursor-origin component; they are structural conditioning variables, not precursor identity labels. All embeddings share the model hidden dimension. Layer normalization precedes memory self-attention, followed by residual attention and feed-forward updates. The resulting program, depth and atom-position tokens provide the memory for cross-attention in the graph denoiser. The morphology fields determine active positions and fixed or variable masks. Sampling from the prior specifies these constraints; generation assigns the atom, bond and topology states within each role.

A chemistry-specific adapter determines the precursor roles, reaction core, reverse decomposition and forward transform associated with c . Its sampling layout determines active positions and fixed or variable coordinates; the embeddings condition neural prediction on the remaining program fields.

A.2 GRAPH SERIALIZATION AND SOURCE ANNOTATIONS

A spanning-tree representation with at most twelve residual closure edges preserves the full 254-heavy-atom support without an $O(N_{\text{max}}^2)$ edge state. The discrete flow generates the atom, bond and topology states in this representation. Let $\alpha = (\mathbf{C}, \tau, \mathbf{r}, \mathbf{k})$ contain the selected exact precursor tuple, registered assembly trace, atom-wise precursor-role assignment and reaction-core annotation. The deterministic program-aware encoder

$$\text{Enc}_P : \{(G, \alpha) : G \in \mathfrak{G}_{N_{\text{max}}, K_{\text{cl}}} \text{ and } \alpha \text{ is admitted for } P\} \longrightarrow \mathfrak{X}_{N_{\text{max}}, K_{\text{cl}}} \quad (13)$$

constructs the categorical graph state

$$X_\star := \text{Enc}_P(G, \alpha) = (\mathbf{a}, \boldsymbol{\pi}, \mathbf{b}, \mathbf{e}^L, \mathbf{e}^R, \mathbf{z}) \quad (14)$$

where $\mathbf{a}$ contains atom states, $\boldsymbol{\pi}$ parent pointers, $\mathbf{b}$ spanning-tree bond states, $\mathbf{e}^L$ and $\mathbf{e}^R$ closure-endpoint pointers, and $\mathbf{z}$ closure-bond states.

The encoder first partitions the atoms into connected precursor-origin components under α . It selects as root the reaction-core atom with the smallest tie-broken canonical RDKit rank, and orders the origin-component graph by breadth-first traversal from that root component, breaking ties by role index, component canonical ranks and original atom index. Within each component it uses a deterministic depth-first preorder from the selected attachment atom, with neighbors ordered by the same canonical ranks. The traversed bonds define the spanning tree. Every residual bond is represented once as an ordered endpoint pair with its bond label, and residual triples are sorted lexicographically. Role, core, fixed-coordinate and morphology arrays follow the same atom order. Each admitted source record supplies a selected qualified exact assembly trace and its precursor-origin assignment. Alternative source-role bindings remain explicit annotations; they do not receive additional graph sampling mass when their product constitutions coincide. Thus $\text{Dec}(\text{Enc}_P(G, \alpha)) \cong G$ for every admitted record.

The raw padded categorical product space $\bar{\mathcal{X}}_{N_{\max}, K_{\text{cl}}}$ also contains malformed states. Its well-formed subset $\mathcal{X}_{N_{\max}, K_{\text{cl}}}$ enforces an active node prefix, earlier-parent pointers, distinct valid closure endpoints, no duplicate edges, and null values in unused closure slots. The total decoder, including its failure value, is typed as

$$\text{Dec} : \bar{\mathcal{X}}_{N_{\max}, K_{\text{cl}}} \longrightarrow \mathcal{G}_{N_{\max}, K_{\text{cl}}} \cup \{\perp_{\text{dec}}\}. \quad (15)$$

For this study, $N_{\max} = 254$ and $K_{\text{cl}} = 12$. The atom vocabulary includes qualified bromine states; bond states encode single, double and triple bonds, with aromatic structures represented through Kekulé bond assignments. These bounds specify architectural support. Reaction consistency, nonzero probability under a trained model and synthesis-program completeness require separate conditions.

The structural decoder specifies graph reconstruction from an already chosen state. The production terminal readout additionally chooses that state from neural predictions (Appendix A.4). These are separate operations.

The chemistry adapter marks any atom, pointer, or bond states fixed by P . FORGE holds these states constant while corrupting the remaining states along the linear probability path and predicting their clean values from (X_t, t, P) . Let Ω_P be the subset of the raw state space satisfying the allowed coordinate alphabets and fixed values, and define the admissible program set

$$\mathcal{P}_{\text{adm}} = \{P : \mathcal{X}_{N_{\max}, K_{\text{cl}}} \cap \Omega_P \neq \emptyset\}. \quad (16)$$

The morphology prior is normalized only on $\mathcal{P}_{\text{adm}}$. Each chemistry adapter is required to satisfy the program-compatible encoding property: every admitted annotated record obeys $\text{Enc}_P(G, \alpha) \in \Omega_P$. Corpus admission requires this property, an exact constitutional encode–decode round trip and source-consistent origin/core assignments. Records failing admission are excluded. The frozen split is [split manifest]; noise marginals, program priors and motif references are fitted only to its training partition. These checks do not establish the property for unregistered chemistry.

A.3 TRANSFORMER ARCHITECTURE AND TRAINING OBJECTIVES

The denoiser is a graph Transformer over the complete sparse state. Each layer applies graph self-attention with relation biases for parent, child and closure edges, then cross-attends to memory tokens representing program identity, program depth, precursor role and reaction-core position. Cross-attention is repeated in every layer so that reaction semantics can influence both local state updates and long-range coordination among precursor-derived regions. The backbone has six layers, hidden dimension 192, eight attention heads and dropout 0.1. Each routed adapter mixes three experts with bottleneck dimension 64 using weights computed from the program summary. Additional routed adapters feed atom/parent-bond predictions and closure predictions. Auxiliary heads predict precursor roles, core positions and exterior child counts. The shared backbone receives repeat-group, component-position and role-morphology conditioning.

Training assigns equal mass to the 22 admitted formal families and uniform mass to qualified unique product constitutions within each family. Sampling is with replacement; multiple source bindings do not multiply a constitution’s probability, and sampling weights are not applied again to the loss. Each optimizer update contains 3,168 examples from three distinct families, with 1,056 examples per selected family. A seeded permutation defines a balanced 22-update cycle in which every family appears three times. Family batches are sharded across eight H100 GPUs, and their coordinate reductions are accumulated before family-gradient projection. Padding is trimmed to the longest molecule within a family batch while retaining the full closure support.

The optimizer is AdamW with learning rate 10^{-4} , weight decay 5×10^{-5} , $(\beta_1, \beta_2) = (0.9, 0.999)$, $\epsilon = 10^{-8}$ and global gradient-norm clipping at 1. The update budget is [optimizer updates]; the independent training seeds and checkpoint-selection rule are [training seeds and checkpoint rule]. Model and gradient computations use deterministic FP32 arithmetic. Checkpoints retain model, optimizer, random-number streams and the committed sampler state; uncommitted prefetch work is discarded on restart. Training times are sampled uniformly, $t \sim \mathcal{U}(0, 1)$, without endpoint clipping.

In the task losses below, c indexes a formal family; multiple registered program variants in that family share its task reduction. For formal family c and state family $q \in \mathcal{J} = \{\mathbf{a}, \pi, \mathbf{b}, \mathbf{e}^L, \mathbf{e}^R, \mathbf{z}\}$, let $\mathcal{S}_{c,q}$ be the selected variable coordinates across that family’s minibatch. With coordinate loss $\ell_i = -\log d_{\theta,i}(X_{\star,i} \mid X_t, t, P)$, the pooled reduction is

$$L_{c,q}^{\text{pool}} = \frac{\sum_{i \in \mathcal{S}_{c,q}} \ell_i}{\max\{1, |\mathcal{S}_{c,q}|\}}. \quad (17)$$

Coordinates here include their example index. Chemistry states (atoms, parent bonds and closure bonds) additionally use equal mass over roles present in the selected coordinates. If $\mathcal{S}_{c,q,r}$ contains coordinates assigned to role r and $\mathcal{R}_{c,q} = \{r : |\mathcal{S}_{c,q,r}| > 0\}$, this reduction is

$$L_{c,q}^{\text{role}} = \frac{1}{|\mathcal{R}_{c,q}|} \sum_{r \in \mathcal{R}_{c,q}} \frac{\sum_{i \in \mathcal{S}_{c,q,r}} \ell_i}{|\mathcal{S}_{c,q,r}|}, \quad (18)$$

with zero loss if no role is present. Atom and parent-bond groups use their position’s role; closure-bond groups use the role at the clean left endpoint. Pointer losses use the pooled reduction and mask logits to the permitted layout positions. The flow term $L_{c,\text{flow}}$ sums these six reduced losses, with unit weights.

For each family, the task loss is

$$L_c = L_{c,\text{flow}} + 0.25L_{c,\text{role}} + 0.25L_{c,\text{core}} + 0.25L_{c,\text{repeat}} + L_{c,\text{child}} + 0.5L_{c,\text{junction}} + L_{c,\text{chem|top}}. \quad (19)$$

The role and core terms are cross-entropies averaged equally over present target states. The repeat term averages squared differences between predicted atom distributions, and separately parent-bond distributions, over exact retained-fragment correspondences in repeated precursor instances; it sums the two means. Atom correspondence requires a unique labelled fragment match that preserves core annotations. Parent-bond correspondence matches both edge endpoints rather than equating serialized child positions. Unequal fragments or ambiguous matches contribute no repeat pairs; these target-derived alignments supervise the loss and are not supplied at inference. The child term is state-balanced cross-entropy for same-component exterior child counts in $\{0, 1, 2, 3\}$. The junction term applies smooth-L1 loss to predicted and target role-local junction totals, each divided by the number of exterior atoms in that role, then averages over eligible examples within a role and equally across present roles. Predicted junction totals sum the expected child-count contribution $\max\{o - 1, 0\}$ across exterior atoms. The topology-conditioned chemistry term repeats the role-balanced atom and bond losses in a second forward pass with clean parent and closure endpoints and corrupted atom and bond states. Each ablation changes only the declared objective or architecture terms under its matched training contract.

PCGrad combines the task gradients in integer formal-family-state order. For $g_c = \nabla_{\theta} L_c$, it initializes $\tilde{g}_c = g_c$ and, for each other task h in that order, applies

$$\tilde{g}_c \leftarrow \tilde{g}_c - \mathbf{1}\{\langle \tilde{g}_c, g_h \rangle < 0\} \frac{\langle \tilde{g}_c, g_h \rangle}{\max\{\|g_h\|_2^2, 10^{-12}\}} g_h \quad (20)$$

on parameter coordinates available to both tasks. Gradients are not norm-normalized, and the optimizer receives the arithmetic mean of the projected task gradients. This update is not generally the gradient of a scalar sum of task losses. No population optimality claim is made for the full procedure. In particular, target-dependent group reductions and the auxiliary objectives require analysis beyond Proposition C.7.

Layerwise cross-attention, auxiliary losses, routed adapters and gradient-conflict control are separate interventions in the architecture comparison (Table 18).

A.4 NUMERICAL SAMPLING AND TERMINAL DECODING

The sampler uses [flow steps] steps, denoted L_{samp} . Its strictly positive atom and bond source marginals are indexed by registered program and precursor role and fitted to TRAIN data. Pointer marginals are restricted and normalized on their allowed layout positions. The finite sampler evaluates the denoiser at $t_k = k/L_{\text{samp}}$, for $k = 0, \dots, L_{\text{samp}} - 1$, and uses $\Delta t = 1/L_{\text{samp}}$. Conditional on $Z_k = x$, it draws one clean category $A_{k,i} \sim d_{\theta,i}(\cdot | x, t_k, P)$ independently for each variable coordinate. Write $s_{t,i}^P(u | a) = (1 - t)q_{0,i}^P(u) + t\mathbf{1}\{u = a\}$ for its conditional path factor, $\dot{s}_i^P(u | a) = \mathbf{1}\{u = a\} - q_{0,i}^P(u)$ for the derivative, and $[z]_+ = \max\{z, 0\}$. The code uses the conditional off-diagonal rate

$$R_{t,i}^{*,\varepsilon}(u, v | a, P) = \frac{[\dot{s}_i^P(v | a) - \dot{s}_i^P(u | a)]_+}{|\mathcal{X}_i(P)| \max\{s_{t,i}^P(u | a), \varepsilon\}}, \quad u \neq v, \quad \varepsilon = 10^{-8}. \quad (21)$$

Full source support is required only on the allowed coordinate alphabet, including the masked candidate set for a pointer. Thus that alphabet is the support for every $t_k < 1$. The denominator clamp in Equation 21 is a numerical safeguard absent from the ideal rate in Appendix C.3.

For the sampled endpoint $a = A_{k,i}$, let $h_{k,i} = \Delta t \sum_{v \neq x_i} R_{t_k,i}^{*,\varepsilon}(x_i, v | a, P)$ and set $\kappa_{k,i} = \min\{1, 0.999/h_{k,i}\}$ when $h_{k,i} > 0$, with $\kappa_{k,i} = 1$ otherwise. The implemented coordinate kernel is

$$\hat{K}_{k,i}(v | x, a) = \begin{cases} \kappa_{k,i} \Delta t R_{t_k,i}^{*,\varepsilon}(x_i, v | a, P), & v \neq x_i, \\ 1 - \kappa_{k,i} h_{k,i}, & v = x_i. \end{cases} \quad (22)$$

Its off-diagonal masses sum to at most 0.999; the diagonal is the remaining mass. Consequently all entries are nonnegative and each row sums to one. Coordinates are drawn independently from these kernels, followed by the projection Π_P .

Without the clamp or cap, and when every conditional Euler row is nonnegative, averaging over $A_{k,i}$ gives the Euler kernel built from Equation 6 exactly. One sampled endpoint per coordinate therefore does not by itself bias that uncapped one-step law. The nonlinear cap generally changes the averaged kernel. Independent coordinate updates also permit simultaneous finite-step changes, so their product is a numerical approximation to the local CTMC, not its exact transition kernel. No discretization error bound is asserted near the terminal singularity.

After all steps, the model evaluates $(Z_{L_{\text{samp}}}, 1, P)$ once more. The production terminal readout H_P^θ selects coordinate values using constrained argmax, restores fixed states and returns either a raw state or a typed terminal failure, denoted collectively by $\perp_{\text{term}}$. The atom-aware readout first chooses atom states subject to unavoidable incident-bond demands from the fixed core and non-root tree attachments; subsequent variable bonds must fit those atom capacities. It abstains when this bounded assignment cannot be completed. It does not exchange an element for a higher-valence element merely to accommodate a chosen topology. The registered core condition reserves the required bond and hydrogen states for each program. In the AGILE-type Ugi 3-CR, these include one hydrogen on the aldehyde-derived α carbon and one on the isocyanide-derived amide nitrogen; these conditions are not imposed on other Ugi variants. The selected state is then reconstructed and checked for connectedness, valence and sanitization. A readout failure retains its detailed reason in the ledger; for graph-law notation we extend $\text{Dec}(\perp_{\text{term}}) = \perp_{\text{dec}}$. If $q_{\theta,\text{out}}^P$ is the raw law after terminal readout, including failure, then

$$p_\theta(\cdot | P) = \text{Dec}_\# q_{\theta,\text{out}}^P, \quad p_\theta(G | P) = \sum_{z: \text{Dec}(z)=G} q_{\theta,\text{out}}^P(z). \quad (23)$$

This law retains structural failures and chemically invalid graphs before the validity and L1 filters; it is not conditioned on acceptance. The ideal theorem does not include the learned $t = 1$ readout, which is unconstrained by a loss integrated over $t \in (0, 1)$. Decoder and source-prior comparisons use their declared fixed-weight or retrained contrasts (Table 9).

A.4.1 CHECKED CONSTRUCTION AND SELECTION

Let Π specify a bounded inference policy: the draw count, ordered proposal operators, TRAIN references, search limits, source verifier, selection objective and selection-group size. Its version is

[inference policy]; the draw count D is [draws per request], the stage caps are [proposal and search caps], and the selection-group size is [selection batch size]. For request b , retain all raw trajectories and form

$$\mathcal{C}_b = \{G_{b,d}^{\text{raw}} : 1 \leq d \leq D\} \cup \mathcal{R}_{\Pi}(P_b, \{Z_{b,d}^{\text{out}}, \lambda_{b,d}\}_{d=1}^D), \quad (24)$$

where $\lambda_{b,d}$ denotes saved terminal predictions and $\mathcal{R}_{\Pi}$ is the bounded construction map. Original graphs, failed trajectories, rejected proposals and exceptions remain in the attempt ledger. A request is not removed when its candidate set has no eligible graph.

Construction can tie repeated components, restore source-required component constraints, propose qualified retained-head sites, and assign TRAIN-supported joint atom/bond or ring-system motifs. The operators act on generated atom slots while preserving the declared atom budget, precursor origins and fixed reaction core. Atom relabelling and topology edits are recorded separately. Ring repair is applied after component and head construction, and all checks are evaluated again on the final graph. Repeated arms receive tied edits when their established correspondences permit this; every proposed product must still pass the unchanged source verifier. A TRAIN motif is an explicit inference prior and does not itself certify chemical plausibility or a complete route. An unsupported or context-dependent ring fragment is unassessed rather than assigned a fabricated known identity. Graph cycle rank, ring systems, SSSR rings and serialization-dependent fundamental cycles remain distinct quantities.

The selector maps the request-indexed pools to at most one graph per request,

$$(G_1^{\text{sel}}, \dots, G_B^{\text{sel}}) = \mathcal{S}_{\Pi}((P_b, \mathcal{C}_b)_{b=1}^B), \quad G_b^{\text{sel}} \in \mathcal{C}_b \cup \{\perp_{\text{dec}}\}. \quad (25)$$

Source-exact eligibility, applicable design predicates and product/role-component diversity and novelty constraints are explicit parts of Π . A combined design pass requires every applicable qualified predicate to be assessed and pass; unassessed checks cannot establish that endpoint. The selection objective and constraint reference are [selector objective and floors]. Each output retains its request, draw and proposal lineage. Raw and selected yields both use the original B requests as denominator, while cost accounting includes BD trajectories, all construction attempts, source checks and selection work.

The selected joint law is the pushforward of the raw trajectory law through $\mathcal{R}_{\Pi}$ and $\mathcal{S}_{\Pi}$. It is distinct from Equation 23 and can couple different requests. The ideal flow theorem does not certify these inference maps; independent-draw accepted-count formulas apply only when their independence assumptions hold. A graph-changing repair changes the proposal distribution, whereas rejecting completed graphs alone does not.

A.5 EXACT ASSEMBLY VERIFICATION

For the chemistry c specified by P , FORGE decomposes each valid graph into role-assigned precursors and applies the registered forward transform. Exact replay checks assembly existence under c ; it does not test full agreement with the sampled conditioning layout.

Definition A.1 (Exact assembly and recovered witnesses). For the registered forward operator $\mathcal{F}_c$ on precursor inputs $\mathcal{C}_c$, define

$$\mathcal{E}_c^*(G) = \{\mathbf{C} \in \mathcal{C}_c : G \in \mathcal{F}_c(\mathbf{C})\}, \quad \text{L1}^*(G, c) = \mathbf{1}\{\mathcal{E}_c^*(G) \neq \emptyset\}. \quad (26)$$

The bounded reverse procedure returns candidate pairs $(\mathbf{C}, \tau)$ in $\mathcal{D}_c(G)$. Equation 2 retains the pairs passing exact forward replay, and

$$\widehat{\text{L1}}(G, c) = \mathbf{1}\{\widehat{\mathcal{E}}_c(G) \neq \emptyset\}. \quad (27)$$

Both predicates describe assembly existence; molecular validity is checked separately.

The set $\mathcal{E}_c^*(G)$ contains precursor inputs, whereas $\widehat{\mathcal{E}}_c(G)$ contains input-trace pairs. Their compatible comparison is

$$\{\mathbf{C} : \exists \tau, (\mathbf{C}, \tau) \in \widehat{\mathcal{E}}_c(G)\} \subseteq \mathcal{E}_c^*(G), \quad \widehat{\text{L1}}(G, c) \leq \text{L1}^*(G, c). \quad (28)$$

The inclusion follows directly from exact replay. Its converse requires complete reverse search, which has not been established. Throughout the paper, verified exact-L1 yield counts valid attempts satisfying the recovered predicate. Candidate traces are compared after the registered role- and step-aware canonicalization, so duplicate encodings do not create additional witnesses.

Post-hoc filtering changes which completed samples are retained but cannot change the proposal distribution’s exact-L1 acceptance probability. Under independent sampling, this probability determines the complete distribution of accepted counts under a fixed attempt budget and the expected cost of obtaining a fixed number of accepted products (Proposition C.10). We distinguish decomposition coverage from forward-replay precision because a candidate disconnection may fail to reconstruct G . Candidate decomposition coverage is the fraction of valid graphs for which $\mathcal{D}_c(G) \neq \emptyset$. Candidate yield additionally uses all original requests as its denominator; undefined conditional ratios are not replaced by zero. For a declared sequence of valid evaluation graphs $G_1, \dots, G_m$, candidate replay precision is

$$\text{ReplayPrecision} = \frac{\sum_{b=1}^m |\hat{\mathcal{E}}_c(G_b)|}{\sum_{b=1}^m |\mathcal{D}_c(G_b)|}, \quad (29)$$

with N/E when the denominator is zero. Replay of tuples already in $\hat{\mathcal{E}}_c(G)$ has 100% success by construction. This invariant does not estimate candidate replay precision. We separately define

$$\text{CandidateAmbig}(G, c) = \mathbf{1}\{|\mathcal{D}_c(G)| > 1\}, \quad \text{ExactAmbig}(G, c) = \mathbf{1}\{|\hat{\mathcal{E}}_c(G)| > 1\}. \quad (30)$$

A distinct exact-L1 product is a deduplicated generated constitution. An unambiguous exact decomposition satisfies $|\hat{\mathcal{E}}_c(G)| = 1$. We distinguish product uniqueness from decomposition uniqueness throughout.

A.6 RECURSIVE ROUTE ASSESSMENT AND DOSSIER ADMISSION

A.6.1 ASSESSMENT INTERFACE

For a valid generated graph G and reaction program P with chemistry c , route assessment returns either a complete synthesis dossier or a typed abstention. Let $\mathfrak{B}$ denote the frozen route contract: depth and query budgets, registered transforms, evidence-admission rules and a dated supplier snapshot. Let $\mathfrak{D}$ be the space of candidate dossiers, each containing a synthesis tree and its reaction and terminal evidence, and let $\mathfrak{F}_{\mathfrak{B}}$ be the disjoint finite set of typed failures. Generation and assessment are defined by

$$Y = \text{Assess}_{\mathfrak{B}}(G, P, \mathbf{C}_T, \tau_T) \in \mathfrak{D} \cup \mathfrak{F}_{\mathfrak{B}}. \quad (31)$$

$\text{Assess}_{\mathfrak{B}}$ is invoked after the verifier returns one admitted root trace $(\mathbf{C}_T, \tau_T) \in \hat{\mathcal{E}}_c(G)$. It verifies the final assembly, searches for preparation routes to unavailable precursors and checks terminal supplier evidence. P and the assessor act at different stages: P guides generation, whereas the assessor constructs a dossier from the completed graph without modifying it. FORGE accepts (G, T) only when G passes molecular and exact-assembly verification and T traces every required precursor to an available starting material; otherwise, it abstains.

A.6.2 RECURSIVE EXPANSION AND COMPLETION CHECKS

Completing a synthesis dossier requires evidence for every recovered precursor. Exact L1 verification specifies the precursor structures; availability and preparation routes require additional evidence. Each precursor must therefore match an available material or have a supported route from available starting materials. Let T be a candidate dossier with bipartite synthesis tree (V_M, V_R, E_T) rooted at G , where each product molecule points to its preparation reaction and each reaction points to its required reactants. The assessor constructs these routes recursively and withholds a complete dossier when any branch remains unresolved. A molecule node terminates when a current, dated supplier record lists an exact constitutional match as available; otherwise, FORGE must expand that node through a supported, forward-verified reaction. FORGE uses a bounded level-two (L2) planner to propose each expansion. For each proposal, FORGE records the reaction, reactants, product, and supporting evidence, and accepts the step only when the forward transform reconstructs the target precursor exactly. Each accepted reaction adds its reactants as new molecule nodes, and FORGE applies the same termination-or-expansion rule recursively. The current assessor admits only an unambiguous verified exact-L1 root. Every dossier therefore records the sole selected root trace

$(\mathbf{C}_T, \tau_T) \in \widehat{\mathcal{E}}_c(G)$; an ambiguous exact set produces a typed abstention. For each upstream reaction node r , let $\text{Admitted}_{\mathfrak{B}}(r) = 1$ when its reaction and substrate scope satisfy the frozen evidence policy, and let $\text{Verify}_{\mathfrak{B}}(r) = 1$ when $\text{target}(r) \in \mathcal{F}_r(\text{reactants}(r))$. For each leaf molecule ℓ , let $\text{Available}_{\mathfrak{B}}(\ell) = 1$ when it satisfies the terminal supplier-evidence rule. Let $V_R^{\text{up}}(T)$ denote the upstream preparation-reaction nodes and $L(T)$ the leaf molecules. $\text{WellFormed}(G, T, P)$ requires that T is a connected, acyclic bipartite tree rooted at G ; each internal molecule has exactly one recorded producing reaction; no molecular constitution repeats along an ancestor path; each reaction’s molecule children equal its recorded reactants with their multiplicities; and T has no disconnected, cyclic or extraneous node. Its distinguished final-assembly record must have product G and precursor children exactly $\mathbf{C}_T$ under trace τ_T .

$$\begin{aligned} \text{Complete}_{\mathfrak{B}}(G, T, P) = & \text{Valid}(G) \text{WellFormed}(G, T, P) \\ & \times \mathbf{1}[\widehat{\mathcal{E}}_c(G) = \{(\mathbf{C}_T, \tau_T)\}] \\ & \times \prod_{r \in V_R^{\text{up}}(T)} \text{Admitted}_{\mathfrak{B}}(r) \text{Verify}_{\mathfrak{B}}(r) \prod_{\ell \in L(T)} \text{Available}_{\mathfrak{B}}(\ell). \end{aligned} \quad (32)$$

All factors are binary and empty products equal one. Repeated use of a material is represented by separate tree occurrences; equality of their constitutions does not merge their nodes. Under finite search bounds and terminating evidence queries, the assessor terminates and returns a dossier only when $\text{Complete}_{\mathfrak{B}}(G, T, P) = 1$ (Lemma C.11 and Invariant C.12). An abstention means that FORGE did not complete a dossier under $\mathfrak{B}$; it does not imply that no synthesis route exists outside the frozen search space or evidence snapshot. The guarantee is computational and does not establish experimental synthesis success.

A.6.3 RECORDED EVIDENCE AND SUPPORT TIERS

For each accepted pair (G, T) , the synthesis dossier contains every molecule and reaction in T , the evidence for each reaction and a dated supplier record for each terminal material. We classify assembly support relative to the fixed reference space into four tiers: **E0 (enumerated)**: the generated lipid matches a product that was already constructed computationally from known components using a supported reaction. **E1 (known components)**: the generated lipid was not included in the bounded E0 enumeration, but a supported reaction assembles it entirely from known components. **E2 (generated precursor)**: a supported reaction assembles the lipid, but at least one precursor is newly generated and requires a complete L2 route to available starting materials. **E3 (new assembly family)**: the lipid requires a final assembly reaction outside the qualified reaction registry.

A.7 END-TO-END INFERENCE PROCEDURE

Algorithm 1 records every requested output and every trajectory used to construct it, including failures in molecular validation, exact replay or route completion.

Algorithm 1 FORGE generation and fail-closed synthesis-program construction.

Require: trained coordinate predictors, chemistry c , program prior $p_{\text{prog}}(\cdot | c)$, integers $L_{\text{samp}}, D \geq 1$ and $B \geq 0$, recorded seed s , inference policy Π , route contract $\mathfrak{B}$

Ensure: attempt-indexed dossier sequence $\mathcal{D}_{\text{out}}$, exact-L1 ledger $\mathcal{L}_{\text{out}}$, and typed failure and proposal ledger $\mathcal{H}$

```

1: initialize the random-number streams from  $s$ ;  $\mathcal{D}_{\text{out}} \leftarrow []$ ;  $\mathcal{L}_{\text{out}} \leftarrow []$ ;  $\mathcal{H} \leftarrow []$ 
2: for  $b = 1, \dots, B$  do
3:   sample and record  $P_b \sim p_{\text{prog}}(\cdot | c)$ ; initialize raw draw ledger  $\mathcal{Z}_b$ 
4:   for  $d = 1, \dots, D$  do
5:     sample  $Z_0 \sim q_0^{P_b}$  and set  $Z \leftarrow \Pi_{P_b}(Z_0)$ 
6:     for  $k = 0, \dots, L_{\text{samp}} - 1$  do
7:        $t_k \leftarrow k/L_{\text{samp}}$ ;  $\Delta t_k \leftarrow 1/L_{\text{samp}}$ 
8:        $Z \leftarrow \Pi_{P_b}(\text{CAPPEDCOORDINATEEULER}(Z, t_k, \Delta t_k, P_b))$ 
9:     end for
10:    obtain terminal predictions  $\lambda_{b,d}$  and  $Z_{b,d}^{\text{out}} \leftarrow H_{P_b}^{\theta}(Z)$ 
11:    append the state or typed terminal failure, predictions and raw decoded graph to  $\mathcal{Z}_b$ 
12:  end for
13:   $\mathcal{C}_b \leftarrow \text{CHECKEDCONSTRUCTION}(P_b, \mathcal{Z}_b, \Pi)$ 
```

```

1080 14:   record every raw/proposed graph, check, abstention and cost in  $\mathcal{H}$ 
1081 15: end for
1082 16:  $(G_1, \dots, G_B) \leftarrow \mathcal{S}_{\Pi}((P_b, \mathcal{C}_b)_{b=1}^B)$ 
1083 17: for  $b = 1, \dots, B$  do
1084 18:    $G \leftarrow G_b; P \leftarrow P_b$ 
1085 19:   if  $G = \perp_{\text{dec}}$  or  $\text{Valid}(G) = 0$  then
1086 20:     record  $(b, \text{INVALIDGRAPH}, \text{decode or validation reason})$  in  $\mathcal{H}$ 
1087 21:     continue
1088 22:   end if
1089 23:    $\hat{\mathcal{E}}_c(G) \leftarrow \{(\mathbf{C}, \tau) \in \mathcal{D}_c(G) : \exists H \in \mathcal{F}_c(\mathbf{C}), H \cong G\}$ 
1090 24:   if  $\hat{\mathcal{E}}_c(G) = \emptyset$  then
1091 25:      $\mathcal{H} \leftarrow \mathcal{H} \parallel [(b, \text{L1FAILURE})]$ 
1092 26:     continue
1093 27:   end if
1094 28:    $\mathcal{L}_{\text{out}} \leftarrow \mathcal{L}_{\text{out}} \parallel [(b, G, \hat{\mathcal{E}}_c(G))]$ 
1095 29:   if  $|\hat{\mathcal{E}}_c(G)| \neq 1$  then
1096 30:      $\mathcal{H} \leftarrow \mathcal{H} \parallel [(b, \text{AMBIGUOUSL1})]$ 
1097 31:     continue
1098 32:   end if
1099 33:   select the sole  $(\mathbf{C}_T, \tau_T) \in \hat{\mathcal{E}}_c(G)$ 
1100 34:    $Y \leftarrow \text{Assess}_{\mathfrak{B}}(G, P, \mathbf{C}_T, \tau_T)$ 
1101 35:   if  $Y$  is a dossier  $T$  and  $\text{Complete}_{\mathfrak{B}}(G, T, P) = 1$  then
1102 36:      $\mathcal{D}_{\text{out}} \leftarrow \mathcal{D}_{\text{out}} \parallel [(b, T)]$ 
1103 37:   else if  $Y \in \mathfrak{F}_{\mathfrak{B}}$  then
1104 38:      $\mathcal{H} \leftarrow \mathcal{H} \parallel [(b, Y)]$ 
1105 39:   else
1106 40:      $\mathcal{H} \leftarrow \mathcal{H} \parallel [(b, \text{INCOMPLETEDOSSIER})]$ 
1107 41:   end if
1108 42: end for
1109 43: return  $\mathcal{D}_{\text{out}}, \mathcal{L}_{\text{out}}, \mathcal{H}$ 

```

A.8 REPRODUCIBILITY AND ARTIFACT PROVENANCE

Each run is identified by its training seed, data and split manifests, atom/program vocabularies, source registries, fitted noise marginals, configuration, code snapshot and checkpoint SHA-256. The independent training-seed identifiers are [\[training seed identifiers\]](#). Each comparison retains its declared attempt budget, checkpoint rule and evaluator. Sampling seeds, request/layout identifiers and proposal lineage are recorded separately from training seeds. The manuscript provenance records identify the source artifacts for tables and figures through SHA-256 hashes.

The inference-policy identifier is [\[inference policy identifier\]](#); its record includes all proposal caps, TRAIN motif references, admission order, selection-group boundaries and source checks. The route contract is [\[route contract identifier\]](#), including depth/query limits, evidence-admission policy and dated terminal-material snapshots. Restarts preserve committed checkpoints and completed request shards, and source-preserving fixes retain the exact earlier implementation and an equivalence receipt. A change to an inference map defines a distinct arm unless equivalence is demonstrated for the declared inputs.

Reproduction requires source code, configuration and environment manifests, model checkpoints, split/registry hashes, attempt-level output and verification ledgers, and figure/table exporters. These artifacts are identified by [\[reproduction archive identifier\]](#). Corpus access conditions are [\[data availability\]](#). Constitutional identity removes atom maps and stereochemical labels while retaining formal charge and bond order; no tautomer normalization is applied.

B EVALUATION ACROSS REACTION FAMILIES

We evaluate complete lipid graphs at three distinct levels: exact assembly consistency, structural and distributional properties, and synthesis-evidence completeness. Unless stated otherwise, N denotes all requested products. Conditional rates specify their assessed population and retain invalid, unsupported and abstained requests in the accompanying accounting.

B.1 COHORT, PROGRAM REGISTRY AND SPLIT ACCOUNTING

We associate each registered assembly program with its source evidence, reactive-site multiplicity, precursor roles and repeat/depth support. Source rows and unique constitutional products are counted separately. The family census includes sampling mass, component inventories, source-study overlap and exclusions; support tiers E0–E3 describe evidence depth rather than a probability of synthesis success.

Table 4: **Cohort census.** Source rows and constitutional products are separate units. Source and role inventories report sampling mass, component counts, source-study overlap and exclusions.

Family	Admitted records	Unique products	Programs / studies	TRAIN / CAL / TEST	Excluded / unassessed
A3 amine–aldehyde–alkyne	—	—	—	—	—
Acid–epoxide diester	—	—	—	—	—
AEMA	—	—	—	—	—
Aldehyde Ugi 3-CR	—	—	—	—	—
Aldehyde Ugi 4-CR	—	—	—	—	—
α -Isocyanoester	—	—	—	—	—
dihydroimidazole	—	—	—	—	—
Amine alkylation	—	—	—	—	—
Amine–epoxide	—	—	—	—	—
Aryl reductive amination	—	—	—	—	—
Aza-Michael acrylamide	—	—	—	—	—
Aza-Michael acrylate	—	—	—	—	—
Disulfide Michael	—	—	—	—	—
Epoxide O-acylation	—	—	—	—	—
iPhos	—	—	—	—	—
Ketone–isocyanide amide	—	—	—	—	—
Ketone Ugi 4-CR	—	—	—	—	—
Maleate	—	—	—	—	—
O-esterification	—	—	—	—	—
Passerini	—	—	—	—	—
Preassembled thiol-yne	—	—	—	—	—
Reductive amination	—	—	—	—	—
Thiolactone	—	—	—	—	—

B.2 TRAINING, NUMERICAL VALIDATION AND COMPUTATIONAL COST

Table 5: **Training and evaluation settings.** Settings and run identifiers are reported separately for each independent training seed.

Field	Setting / result
Cohort, split, source registry and vocabulary hashes	—
Checkpoint IDs, independent seed IDs and stopping rule	—
Per-family sampling law, presentations and update appearances	—
Global/per-device batch, accumulation and optimizer updates	—
Optimizer, learning-rate schedule, loss weights and clipping	—
Architecture, adapters, role/core/repeat and topology objectives	—
Precision, determinism, devices, workers and memory	—
Numerical equivalence, kill/restart and checkpoint provenance	—
Training, setup, checkpoint, sampling and verifier time/cost	—
Requests per family/seed, saved draw IDs and final inference policy	—
Sampled-layout distribution and all method-visible information	—

Training diagnostics resolve atom, bond, parent, closure and repeated-component errors by family and corruption time. Checkpoint comparisons use fixed development requests and report exact yield, structural assessment and diversity alongside gradients and clipping. We distinguish molecular presentations, family appearances in updates and optimizer updates. Matched-exposure batch/update comparisons and exposure-scaling comparisons isolate different effects. Checkpoint selection uses [\[checkpoint-selection rule\]](#).

a. Learning and exposure

Per-family losses and errors against presentations and optimizer updates.

[\[result\]](#)

b. Quality and cost

Fixed development checkpoint metrics; latency/calls per accepted distinct product.

[\[result\]](#)

Figure 2: **Training adequacy and computational efficiency.** Learning curves distinguish presentations from optimizer updates. Hardware, precision, deterministic settings, numerical failures and elapsed budget are reported for each arm.

B.3 EXACT-L1 DECOMPOSITION AND FAILURE ACCOUNTING

We report exact assembly outcomes for each family, registered program, training seed and inference arm. Each request contributes to the denominator whether generation, decomposition or verification succeeds. Alongside exact yield, we count valid and connected graphs, distinct exact products, unique exact precursor roots and checked decomposition traces. Candidate-search coverage is the fraction of requests for which a decomposition candidate is found; verified decomposition coverage requires exact replay. Candidate precision counts passing checked traces over all checked traces, with ambiguity and search censoring explicit. The replay rate among already accepted exact traces is an implementation invariant, not an independent precision estimate. We distinguish family-local product counts from counts deduplicated across all families. Equal-family macro averages, pooled yields and worst-family outcomes use separate definitions. Training-seed variability accompanies the family-wise estimates.

Table 6: **Attempt-level outcomes.** Raw and selected outputs are evaluated separately for each registered program and seed. Malformed graphs, support violations, absent decompositions, rejected source domains, exceptions and search censoring are separate outcomes. Counts retain all-request denominators; nonexclusive events may overlap.

Family	Valid / requests	Connected / requests	Distinct exact / N	Ambiguous exact roots	Failure / abstention
A3 amine–aldehyde–alkyne	—	—	—	—	—
Acid–epoxide diester	—	—	—	—	—
AEMA	—	—	—	—	—
Aldehyde Ugi 3-CR	—	—	—	—	—
Aldehyde Ugi 4-CR	—	—	—	—	—
α -Isocyanoester	—	—	—	—	—
dihydroimidazole	—	—	—	—	—
Amine alkylation	—	—	—	—	—
Amine–epoxide	—	—	—	—	—
Aryl reductive amination	—	—	—	—	—
Aza-Michael acrylamide	—	—	—	—	—
Aza-Michael acrylate	—	—	—	—	—
Disulfide Michael	—	—	—	—	—
Epoxide O-acylation	—	—	—	—	—
iPhos	—	—	—	—	—
Ketone–isocyanide amide	—	—	—	—	—
Ketone Ugi 4-CR	—	—	—	—	—
Maleate	—	—	—	—	—
O-esterification	—	—	—	—	—
Passerini	—	—	—	—	—
Preassembled thiol-yne	—	—	—	—	—
Reductive amination	—	—	—	—	—
Thiolactone	—	—	—	—	—

Table 7: **Independent training-seed outcomes.** Each column denotes an independent training run, with identifiers [training seed IDs]. We report sample counts, paired differences and observed ranges for each method and inference arm. Sampling noise from one checkpoint does not constitute an independent training seed.

Family	Seed A exact / N	Seed B exact / N	Seed C exact / N	Mean $\pm$ sample SD	Paired control contrasts
A3 amine–aldehyde–alkyne	—	—	—	—	—
Acid–epoxide diester	—	—	—	—	—
AEMA	—	—	—	—	—
Aldehyde Ugi 3-CR	—	—	—	—	—
Aldehyde Ugi 4-CR	—	—	—	—	—
α -Isocyanoester	—	—	—	—	—
dihydroimidazole	—	—	—	—	—
Amine alkylation	—	—	—	—	—
Amine–epoxide	—	—	—	—	—
Aryl reductive amination	—	—	—	—	—
Aza-Michael acrylamide	—	—	—	—	—
Aza-Michael acrylate	—	—	—	—	—
Disulfide Michael	—	—	—	—	—
Epoxide O-acylation	—	—	—	—	—
iPhos	—	—	—	—	—
Ketone–isocyanide amide	—	—	—	—	—
Ketone Ugi 4-CR	—	—	—	—	—
Maleate	—	—	—	—	—
O-esterification	—	—	—	—	—
Passerini	—	—	—	—	—
Preamsembled thiol-yne	—	—	—	—	—
Reductive amination	—	—	—	—	—
Thiolactone	—	—	—	—	—

B.4 MATCHED CONTROLS, NATIVE BASELINES AND THE THREE-FAMILY COMPARISON

Table 8: **Matched controls and three-family comparison.** Each comparison specifies compatible programs, splits, attempt budgets and verification. Unsupported native chemistry is unassessed, not a zero score.

Arm / comparison	Exact L1	Distinct novel	Quality axes	Cost
Conditioned shared model	—	—	—	—
Independently trained shared-null/post-hoc	—	—	—	—
Independently trained cyclic-label control	—	—	—	—
Finite TRAIN catalogue and learned selector	—	—	—	—
FACT, matched and generous capacity	—	—	—	—
DeFoG / GenMol-SAFE / RGFN, qualified scope	—	—	—	—
Three-family versus 22-family training, common programs	—	—	—	—

Control comparisons use paired request distributions and report differences within each training seed, observed ranges and uncertainty. Training exposure, checkpoint selection, decoder opportunity and source/planner budgets define the matched comparison. We distinguish supported native chemistry from unavailable baseline coverage. A wrong reaction label supplied alongside correct roles and reaction cores tests the incremental information in that label; a retrained null model additionally removes the corresponding training signal.

B.5 DECODER AND STRUCTURAL-REPAIR ATTRIBUTION

Table 9: **Fixed-weight inference ablations.** Original, proposed and selected graphs share the same saved requests and model logits. Larger candidate-pool comparisons are separated from matched-cost controls; fixed-weight changes measure inference contributions.

Inference arm	Exact L1	Design checks	Diversity	Added cost
Raw readout	—	—	—	—
Constrained topology / core readout	—	—	—	—
Checked component completion	—	—	—	—
Scoped head-survival proposals	—	—	—	—
Joint role atom/bond motif proposals	—	—	—	—
Final ring-system topology proposals	—	—	—	—
Combined repairs, original candidates retained	—	—	—	—
Combined repairs + declared selection rule	—	—	—	—
Equal-cost control / no TRAIN-motif prior	—	—	—	—

The inference analysis separates readout, component completion, local chemical proposals, ring-system proposals and population selection. We retain original, proposed and selected graphs, with atom, bond and topology edits, repeated-arm identities, rejected proposals and search limits. Every design predicate is evaluated on the final graph because a later construction can undo an earlier correction. Cost includes duplicate and failed proposals, source checks and selection. TRAIN local motifs used as proposal priors are identified separately from independent evaluation references.

B.6 STRUCTURAL PLAUSIBILITY AND DESIGN-CONDITION AGREEMENT

Table 10: **Structural assessment by family.** All-request outcomes are accompanied by a separate matched source-positive control population. Each rule has applicable, assessed, pass, fail and abstain counts; head and ring agreement are distinct endpoints. Exact-plus-design yield requires every applicable check to be assessed and pass; abstentions and unknowns cannot enter its numerator, and all requests remain in its denominator; it is not a universal quality score. Context flags are not automatic chemical failures. Control disagreements and sensitivity limits define the scope of interpretation.

Family	Exact + design pass / <i>N</i>	Represent. failures	Chemical alerts	Context flags	Head / ring agreement	Abstain / unassessed
A3 amine–aldehyde–alkyne	—	—	—	—	—	—
Acid–epoxide diester	—	—	—	—	—	—
AEMA	—	—	—	—	—	—
Aldehyde Ugi 3-CR	—	—	—	—	—	—
Aldehyde Ugi 4-CR	—	—	—	—	—	—
α -Isocyanoester	—	—	—	—	—	—
dihydroimidazole	—	—	—	—	—	—
Amine alkylation	—	—	—	—	—	—
Amine–epoxide	—	—	—	—	—	—
Aryl reductive amination	—	—	—	—	—	—
Aza-Michael acrylamide	—	—	—	—	—	—
Aza-Michael acrylate	—	—	—	—	—	—
Disulfide Michael	—	—	—	—	—	—
Epoxide O-acylation	—	—	—	—	—	—
iPhos	—	—	—	—	—	—
Ketone–isocyanide amide	—	—	—	—	—	—
Ketone Ugi 4-CR	—	—	—	—	—	—
Maleate	—	—	—	—	—	—
O-esterification	—	—	—	—	—	—
Passerini	—	—	—	—	—	—
Preassembled thiol-yne	—	—	—	—	—	—
Reductive amination	—	—	—	—	—	—
Thiolactone	—	—	—	—	—	—
Equal-family macro	—	—	—	—	—	—
Pooled all requests	—	—	—	—	—	—
Worst-family result	—	—	—	—	—	—

Structural assessment records sanitization, connectedness, radical/charge/vocabulary support, chemical alerts, retained head sites, carbon-domain sizes and branching, head-to-domain distances and connected ring systems. Alert definitions are fixed before candidate review. No universal tail-length, logP or pKa cutoff is imposed. Graph cycle rank and connected ring systems are distinct from SSSR rings and serialization-dependent fundamental cycles; incompatible basis comparisons abstain. Visual assessment uses a method-masked sample from every family and matched controls, with sampling seed [visual sampling seed] and reviewers [reviewer type]. Graph-specific reasons, uncertainty and depiction corrections accompany the review labels.

Table 11: **Head, ring and TRAIN-support diagnostics by family.** Head retention uses the declared family-scope request denominator, including charged/ambiguous abstentions. Cycle allocation uses all requests, including matching zero closures; variable ring-size agreement uses comparable cyclic requests, with unassessed counts. Atom coverage averages assessed products. Exact-plus-observed requires product and all component features, divided by all requests. TRAIN support is descriptive and never substitutes for independent quality evidence.

Family	Head retention	Role cycle allocation	Ring-size agreement	Atom-env. coverage	Exact + observed product/comps. / <i>N</i>
A3 amine–aldehyde–alkyne	—	—	—	—	—
Acid–epoxide diester	—	—	—	—	—
AEMA	—	—	—	—	—
Aldehyde Ugi 3-CR	—	—	—	—	—
Aldehyde Ugi 4-CR	—	—	—	—	—
α -Isocyanoester	—	—	—	—	—
dihydroimidazole	—	—	—	—	—
Amine alkylation	—	—	—	—	—
Amine–epoxide	—	—	—	—	—
Aryl reductive amination	—	—	—	—	—
Aza-Michael acrylamide	—	—	—	—	—
Aza-Michael acrylate	—	—	—	—	—
Disulfide Michael	—	—	—	—	—
Epoxide O-acylation	—	—	—	—	—
iPhos	—	—	—	—	—
Ketone–isocyanide amide	—	—	—	—	—
Ketone Ugi 4-CR	—	—	—	—	—
Maleate	—	—	—	—	—
O-esterification	—	—	—	—	—
Passerini	—	—	—	—	—
Preassembled thiol-yne	—	—	—	—	—
Reductive amination	—	—	—	—	—
Thiolactone	—	—	—	—	—

B.7 LIPID REALISM INDEPENDENT OF DECODER PRIORS

Table 12: **Reference-distribution assessment.** We compare source-held-out empirical lipids and family-matched controls as separate reference populations. Descriptor distance is reported feature-wise, including tails, not collapsed to an arbitrary acceptability cutoff. Valid and unique counts, group splits, uncertainty and missing reference support accompany each comparison.

Family	Fingerprint prec. / cov.	Descriptor prec. / cov.	Grouped C2ST AUC	Descriptor distance	Reference / generated n
A3 amine–aldehyde–alkyne	—	—	—	—	—
Acid–epoxide diester	—	—	—	—	—
AEMA	—	—	—	—	—
Aldehyde Ugi 3-CR	—	—	—	—	—
Aldehyde Ugi 4-CR	—	—	—	—	—
α -Isocyanoester	—	—	—	—	—
dihydroimidazole	—	—	—	—	—
Amine alkylation	—	—	—	—	—
Amine–epoxide	—	—	—	—	—
Aryl reductive amination	—	—	—	—	—
Aza-Michael acrylamide	—	—	—	—	—
Aza-Michael acrylate	—	—	—	—	—
Disulfide Michael	—	—	—	—	—
Epoxide O-acylation	—	—	—	—	—
iPhos	—	—	—	—	—
Ketone–isocyanide amide	—	—	—	—	—
Ketone Ugi 4-CR	—	—	—	—	—
Maleate	—	—	—	—	—
O-esterification	—	—	—	—	—
Passerini	—	—	—	—	—
Preassembled thiol-yne	—	—	—	—	—
Reductive amination	—	—	—	—	—
Thiolactone	—	—	—	—	—
Equal-family macro	—	—	—	—	—
Pooled all requests	—	—	—	—	—
Worst-family result	—	—	—	—	—

Aggregation. AUC and distribution distances are not pooled success ratios. We report equal-family macro means, recomputed pooled estimates and worst-family values separately, with the direction of each metric stated. Unsupported reference populations remain unassessed; missing-family counts accompany each aggregate.

Descriptor comparisons cover molecular size, lipophilicity, polar surface area, hydrogen-bond counts, flexibility, rings and aromaticity, charge, heteroatom composition, branching, unsaturation and graph-based lipid regions. We report feature-wise reference distributions and normalized Wasserstein distances. Fingerprint precision and coverage, source-grouped classifier AUC and TRAIN feature membership quantify different properties. Source-grouped splits exclude duplicate identities across classifier partitions and do not supply source identifiers as features. No descriptor or similarity metric establishes biological utility.

B.8 EXACT DEFINITIONS OF THE QUALITY AND REALISM METRICS

The metrics below distinguish structural conditions, reference membership and distributional similarity. Reference populations, constitutional identity, implementations and random seeds are fixed for each comparison. A metric’s definition does not imply that a generated population satisfies it.

Table 13: **Quality and realism metric definitions.** Each metric has an explicit numerator, denominator and interpretation. Structural checks and reference-support metrics are not interchangeable.

Metric	Exact computation / limitation
Valid molecules / all requests (%)	RDKit parsing/sanitization; failure remains in all-request denominator.
Connected molecules / all requests (%)	Exactly one connected molecular graph; distinct from validity.
Exact L1 + all qualified design checks / requests (%)	Source-exact final graph and every applicable qualified axis assessed/pass. Unknowns and abstentions do not pass.
Molecules with a justified chemical alert / requests (%)	Frozen narrow chemical-alert motifs, with per-rule counts and unassessed requests reported separately. This is not a complete stability screen.
Molecules requiring chemistry-context review / requests (%)	Frozen context flags are descriptive, not automatic failures.
Required head retained / family-scope requests (%)	Pass / prespecified family-scope requests, including charged or ambiguous-origin abstentions. Only neutral aldehyde-Ugi4 has a qualified predicate; other families remain unassessed.
Correct role-local cycle allocation / requests (%)	Match declared variable closures per role, including zero; cross-origin edges must also match. All requests remain in the denominator; unassessed cases are reported separately.
Requested ring-size agreement / comparable requests (%)	Each observed fundamental-cycle length must belong to its role’s allowed-size set, under a comparable tree basis. Not a size-multiset, SSSR or 3D-geometry match. Inapplicable and unassessed counts are reported separately.
TRAIN atom-environment coverage (%)	Fraction of atoms with reference-observed local environments, averaged over assessed products. Component-role summaries are reported separately.
Ring-system TRAIN support / assessed products (%)	Among valid connected products with a family TRAIN reference, fraction with no unknown connected ring-bond signature. Ring-free products pass vacuously; their counts and missing/invalid cases are reported separately. Component roles are separate.
Exact L1 + product/component features observed / requests (%)	Exact, unambiguously decomposed product and every precursor must have all atom environments and connected ring-bond signatures observed in their family product or family/source-role TRAIN reference. The denominator is all requests; absent references never pass.
Fingerprint precision (%)	Fraction of unique generated molecules inside the union of reference k-nearest-neighbor balls in ECFP4/Tanimoto distance.
Fingerprint coverage (%)	Fraction of reference-centered neighborhoods hit by at least one unique generated molecule.
Fingerprint recall (%)	Fraction of references inside generated-centered neighborhoods. Distinct from coverage; small generated sets can inflate neighborhood radii.
Distinct reference-supported products / requests (%)	Unique generated identities inside reference neighborhoods divided by all requests; exposes invalidity and duplication.
Nearest-reference Tanimoto similarity	Mean nearest-reference similarity over unique generated identities; descriptive, not a novelty or quality objective.
Internal ECFP4 diversity	Mean Tanimoto distance between distinct generated identities, interpreted alongside validity, realism and component diversity.
Descriptor distribution distance, per feature	One-dimensional Wasserstein distance / reference IQR; near-zero fallback: max(reference population SD, 1). Connected-request observations retain duplicates. Smaller means closer to reference, not better chemistry.
Descriptor-space precision and coverage	Precision measures generated points in reference-centered descriptor neighborhoods; coverage measures reference neighborhoods reached by generated points. Scaling and estimator: [descriptor neighborhood estimator] ; source-held-out reference: [reference cohort] .

Source-grouped two-sample classifier AUC AUC distinguishes generated from reference molecules across held-out source groups. Classifier and sampling: [C2ST estimator]; grouping and duplicate controls: [C2ST split].

Table 14: **Lipid descriptor distributions.** The 24 descriptors are compared feature by feature on connected-request observations, retaining duplicates. We report means, SDs and distribution tails separately from model-seed uncertainty, alongside normalized Wasserstein distance. W_1/s_R denotes reference-scaled one-dimensional Wasserstein distance. Each family/reference cohort has a separate panel; no arbitrary acceptance threshold is imposed.

Descriptor	Unit	Reference	Raw	Final	W_1/s_R
Heavy atoms	count	—	—	—	—
Molecular weight	Da	—	—	—	—
Calculated logP	unitless	—	—	—	—
Topological polar surface area	Å ²	—	—	—	—
Hydrogen-bond donors	count	—	—	—	—
Hydrogen-bond acceptors	count	—	—	—	—
Rotatable bonds	count	—	—	—	—
Rings	count	—	—	—	—
Aromatic rings	count	—	—	—	—
Fraction sp ³ carbons	fraction	—	—	—	—
Net formal charge	elementary charge	—	—	—	—
Sum of absolute formal charges	elementary charge	—	—	—	—
Carbon fraction	fraction	—	—	—	—
Nitrogen atoms	count	—	—	—	—
Oxygen atoms	count	—	—	—	—
Sulfur atoms	count	—	—	—	—
Phosphorus atoms	count	—	—	—	—
Branching heavy atoms	count	—	—	—	—
Nonaromatic unsaturated bonds	count	—	—	—	—
Maximum polar-root distance	graph bonds	—	—	—	—
Heuristic head fraction	fraction	—	—	—	—
Heuristic interface fraction	fraction	—	—	—	—
Heuristic tail fraction	fraction	—	—	—	—
Heuristic tail branching atoms	count	—	—	—	—

Head/interface/tail fractions and polar-root distance use the descriptor implementation’s graph heuristic; they are not precursor-origin annotations or an independently qualified headgroup test. Branching heavy atoms use heavy-atom degree at least three; carbon-domain branching in the structural audit is a separate measurement. Formal charge is not ionization at physiological pH.

B.9 PRODUCT AND PRECURSOR DIVERSITY, NOVELTY AND REACHABILITY

Table 15: **Product and component diversity and novelty.** N is all requests. D_1 and D_2 are Shannon and inverse-Simpson effective counts, reported per registered role. Novel-component endpoints require an unambiguous exact decomposition and include separate observation and distinct-product yields. TRAIN and full-declared-reference novelty are separate endpoints.

Family	Distinct exact / N	Novel product / N	Novel comp. product / N	Component D_1 / D_2	Novel distinct components
A3 amine–aldehyde–alkyne	—	—	—	—	—
Acid–epoxide diester	—	—	—	—	—
AEMA	—	—	—	—	—
Aldehyde Ugi 3-CR	—	—	—	—	—
Aldehyde Ugi 4-CR	—	—	—	—	—
α -Isocyanoester	—	—	—	—	—
dihydroimidazole	—	—	—	—	—
Amine alkylation	—	—	—	—	—
Amine–epoxide	—	—	—	—	—
Aryl reductive amination	—	—	—	—	—
Aza-Michael acrylamide	—	—	—	—	—
Aza-Michael acrylate	—	—	—	—	—
Disulfide Michael	—	—	—	—	—
Epoxide O-acylation	—	—	—	—	—
iPhos	—	—	—	—	—
Ketone–isocyanide amide	—	—	—	—	—
Ketone Ugi 4-CR	—	—	—	—	—
Maleate	—	—	—	—	—
O-esterification	—	—	—	—	—
Passerini	—	—	—	—	—
Preassembled thiol-yne	—	—	—	—	—
Reductive amination	—	—	—	—	—
Thiolactone	—	—	—	—	—
Equal-family macro	—	—	—	—	—
Pooled all requests	—	—	—	—	—
Worst-family result	—	—	—	—	—

Product and per-role component distributions are summarized by unique counts, frequency tables, largest-identity concentration and effective diversity. Novelty is evaluated separately against TRAIN and the full declared reference. Absolute noninferiority comparisons preserve the baseline observation denominators as well as unique and effective counts, including novel observations and distinct novel identities. Global unique counts deduplicate across families. Catalogue inputs, admissible precursor tuples and reachable unique products are separate quantities: a novel accepted precursor does not by itself show that the product is unreachable through every catalogue decomposition.

B.10 GENERALIZATION BEYOND DEVELOPMENT REQUESTS

Table 16: **Generalization and held-component recovery.** Each split has its own request denominator and independent training-seed outcomes. Unsupported cases remain unassessed. Size, closure and depth strata are reported separately. Zero-shot held-reaction-family behavior is a secondary stress test, not a hard gate.

Family	Held comp. yield / N	Held identity coverage	Component- disjoint L1	Source- disjoint L1	Size/repeat strata
A3 amine–aldehyde–alkyne	—	—	—	—	—
Acid–epoxide diester	—	—	—	—	—
AEMA	—	—	—	—	—
Aldehyde Ugi 3-CR	—	—	—	—	—
Aldehyde Ugi 4-CR	—	—	—	—	—
α -Isocyanoester	—	—	—	—	—
dihydroimidazole	—	—	—	—	—
Amine alkylation	—	—	—	—	—
Amine–epoxide	—	—	—	—	—
Aryl reductive amination	—	—	—	—	—
Aza-Michael acrylamide	—	—	—	—	—
Aza-Michael acrylate	—	—	—	—	—
Disulfide Michael	—	—	—	—	—
Epoxide O-acylation	—	—	—	—	—
iPhos	—	—	—	—	—
Ketone–isocyanide amide	—	—	—	—	—
Ketone Ugi 4-CR	—	—	—	—	—
Maleate	—	—	—	—	—
O-esterification	—	—	—	—	—
Passerini	—	—	—	—	—
Preamsembled thiol-yne	—	—	—	—	—
Reductive amination	—	—	—	—	—
Thiolactone	—	—	—	—	—
Equal-family macro	—	—	—	—	—
Pooled all requests	—	—	—	—	—
Worst-family result	—	—	—	—	—

Model, decoder and selection choices are fixed on development data before evaluation on the declared held-out splits. We distinguish independent training seeds from sampling seeds and report both request/layout identity overlap and graph-geometry overlap. Fresh noise on TRAIN-derived layouts measures sampling variability, not held-out generalization. Uncertainty uses [population-aware uncertainty estimator] to account for dependence introduced by joint population selection.

B.11 UPSTREAM SYNTHESIS AND TERMINAL-EVIDENCE COMPLETENESS

Table 17: **L1/L2/L3 evidence and dossier dispositions.** Products, unique family-role component constitutions, reaction steps and terminal leaves are distinct denominators. Complete dossiers require all admitted steps and leaves for the actual generated identity. L2/L3 unknowns, source conflicts, search limits and access failures are counted separately.

Family	Exact L1 products	L2 closed / branches	L3 qualified / leaves	Complete dossiers / <i>N</i>	Unresolved / censored
A3 amine-aldehyde-alkyne	—	—	—	—	—
Acid-epoxide diester	—	—	—	—	—
AEMA	—	—	—	—	—
Aldehyde Ugi 3-CR	—	—	—	—	—
Aldehyde Ugi 4-CR	—	—	—	—	—
α -Isocyanoester	—	—	—	—	—
dihydroimidazole	—	—	—	—	—
Amine alkylation	—	—	—	—	—
Amine-epoxide	—	—	—	—	—
Aryl reductive amination	—	—	—	—	—
Aza-Michael acrylamide	—	—	—	—	—
Aza-Michael acrylate	—	—	—	—	—
Disulfide Michael	—	—	—	—	—
Epoxide O-acylation	—	—	—	—	—
iPhos	—	—	—	—	—
Ketone-isocyanide amide	—	—	—	—	—
Ketone Ugi 4-CR	—	—	—	—	—
Maleate	—	—	—	—	—
O-esterification	—	—	—	—	—
Passerini	—	—	—	—	—
Preassembled thiol-yne	—	—	—	—	—
Reductive amination	—	—	—	—	—
Thiolactone	—	—	—	—	—
Equal-family macro	—	—	—	—	—
Pooled all requests	—	—	—	—	—
Worst-family result	—	—	—	—	—

Route assessment follows the actual generated precursor identity. Method-masked adjudication of the component union records source identifiers and locators, source/file hashes, reactive-site and forward checks, terminal identity and availability dates, route depths, planner calls and elapsed time. We report E0–E3 dispositions and distinguish raw from selection-conditioned coverage. An E2 assignment requires closure of the routes supporting its generated precursors. Evidence depth is family-specific; detailed Ugi evidence is not transferred to another reaction family. L3 denotes documentary terminal-material qualification rather than prospective procurement or synthesis.

B.12 ARCHITECTURE, THEORETICAL ASSUMPTIONS AND SYNTHESIS GUIDANCE

Table 18: **Architecture comparisons.** Comparisons use matched retraining and report training exposure, capacity and inference cost separately.

Contrast / readouts	Result
Whole graph versus role-factorized, matched/generous capacity	—
Input-only versus layerwise program conditioning	—
Role/core, repeat, topology and chemistry-loss contributions	—
Routed adapters/heads and gradient-conflict handling	—
Per-family exact yield, held losses, diversity and dependence	—

Table 19: Correspondence between theory and implementation.

Obligation	Evidence
Bounded vocabularies/size/closures and actual production masks	—
Reference objective versus implemented training reductions/losses	—
Numerical sampler, endpoint readout, repair and selection maps	—
Preserved invariants and properties requiring empirical evidence	—
Independent-draw assumptions versus batch-coupled selection	—

Table 20: **Synthesis-guidance comparisons.** We distinguish synthesis-guidance effects from a separately scoped potency diagnostic; all comparisons retain eligibility, abstention and budget accounting.

Validation / contrast	Result
Nontrivial generated L1/L2/L3 dossiers and zero-guidance identity	—
Matched unguided, post-hoc and in-trajectory synthesis contrast	—
Eligibility, abstention, route yield, diversity and total budget	—
Potency-only diagnostic with separate applicability policy	—

Architecture comparisons distinguish matched retraining from fixed-weight inference changes. We relate each theoretical assumption to the implemented vocabulary, graph support, objective, finite-step sampler and selection map. The ideal flow theorem does not certify a learned numerical sampler, a TRAIN-derived repair or a batch-coupled selection procedure. Synthesis-guidance comparisons separate unguided, post-hoc and in-trajectory arms at matched total budgets and require both a zero-guidance identity check and auditable L1/L2/L3 dossiers. Outcomes outside the declared evidence or applicability scope remain unassessed.

B.13 ASSEMBLY, STRUCTURAL AND SYNTHESIS-EVIDENCE OUTCOMES

a. Assembly and conditions Per-family raw and selected exact yield, request counts and training-seed variability. [result]	b. Structural assessment Chemical alerts, qualified condition failures and explicit abstentions. [result]
c. Diversity and realism Product and role diversity, novelty and reference precision/coverage. [result]	d. Synthesis evidence L1, L2 and L3 dispositions and complete-dossier yield per request. [result]

Figure 3: **Assembly, molecular properties and synthesis evidence across reaction families.** Assembly consistency, structural assessment, distributional similarity and evidence depth are separate outcomes. Family-wise estimates retain their own denominators, comparator budgets and training-seed uncertainty. Raw and selected populations are distinguished throughout.

B.14 REPRESENTATIVE STRUCTURES AND UNRESOLVED FAILURES

a. Seeded all-family atlas

One or more prespecified random products per family, with exact precursors and evidence depth.

[result]

b. Paired failure analysis

Original and repaired graphs; edits, remaining concerns and unchanged source checks.

[result]

Figure 4: **Structures and failure analysis across reaction families.** The atlas uses a fixed selection seed across all families. Atom origins, component novelty and L1/L2/L3 evidence are annotated separately. Random samples and targeted failure examples are distinguished. Molecular drawings represent exact graphs and do not establish three-dimensional stability.

The structure analysis pairs original and repaired graphs and includes oversized rings, malformed cages, missing retained heads and joint atom/bond errors, alongside source-positive controls. Graph and drawing hashes distinguish molecular changes from depiction corrections. Random samples and examples selected to illustrate a specific failure are identified separately.

B.15 FAILURE ACCOUNTING AND REPRODUCIBILITY

The analysis retains failed requests and runs, unsupported chemistry, unavailable references, source conflicts, null contrasts and search censoring. Source, configuration, checkpoint, policy and split hashes link each result to its executable analysis and verification receipt. We distinguish TRAIN-derived proposal and selection diagnostics from evaluation against independent references. Passing limited structural checks does not establish complete chemical quality, stability, ionization or delivery activity; context flags and unresolved cases remain part of the reported outcome.

C THEORETICAL ANALYSIS AND PROOFS

The results establish separate properties of the construction. Bounded completeness concerns the state representation. Catalogue reachability concerns exact precursor inputs. The population argument connects a reference denoising loss to an ideal continuous-time law. Budget and route results concern computational procedures. None establishes experimental synthesis success or positive learned probability for every representable graph.

C.1 GRAPH STATES AND PROGRAM COMPATIBILITY

The graph space $\mathfrak{G}_{N,K_{\text{cl}}}$ is defined in Section 2.3, with finite atom and bond alphabets and integers $N \geq 1, K_{\text{cl}} \geq 0$. Molecular validity is an additional predicate on that space. The raw space $\bar{\mathfrak{X}}_{N,K_{\text{cl}}}$ contains padded categorical states.

Definition C.1 (Well-formed sparse state). A state belongs to $\bar{\mathfrak{X}}_{N,K_{\text{cl}}}$ when, for some $1 \leq n \leq N$, its active atoms are exactly positions $1, \dots, n$ with labels in Σ_{atom} ; position 1 is the sole root with null parent and parent-bond states; and every $2 \leq i \leq n$ has an earlier parent $1 \leq \pi_i < i$ and a bond label in Σ_{bond} . Each active closure has endpoints $1 \leq e^L < e^R \leq n$, a bond label in Σ_{bond} , and an edge distinct from every tree edge and other closure. Active closures precede unused slots and are sorted by endpoint pair. All unused atom, pointer and bond coordinates carry their designated null values.

The decoder reconstructs the labeled edges of a well-formed state and returns $\perp_{\text{dec}}$ otherwise. It is therefore a total map into $\mathfrak{G}_{N,K_{\text{cl}}} \cup \{\perp_{\text{dec}}\}$.

Lemma C.2 (Tree residual count). *For a finite nonempty connected graph $G = (V, E)$ and any spanning tree T , $|E \setminus E(T)| = |E| - |V| + 1 = \beta(G)$.*

Proof. The tree contains $|V| - 1$ edges, all in E . Subtracting this count from $|E|$ gives the result, including the single-vertex case. $\square$

Theorem C.3 (Bounded completeness of the sparse representation). *For the spaces in Definition C.1, the restriction $\text{Dec} : \bar{\mathfrak{X}}_{N,K_{\text{cl}}} \rightarrow \mathfrak{G}_{N,K_{\text{cl}}}$ is well defined and surjective. Thus every well-formed state decodes to a graph in the declared support, and every graph in that support has an encoding that decodes to its constitutional isomorphism class.*

Proof. We first reconstruct a graph from a state, then construct a state for an arbitrary graph. In a well-formed state with n active atoms, every parent pointer strictly decreases its index. Following parents must reach position 1, so the $n - 1$ parent edges form a connected tree. Adding $k \leq K_{\text{cl}}$ distinct, non-loop closure edges preserves connectedness and gives cycle rank $(n - 1 + k) - n + 1 = k$. The label and padding conditions ensure the decoded graph belongs to $\mathfrak{G}_{N,K_{\text{cl}}}$.

Conversely, choose a representative of any $G \in \mathfrak{G}_{N,K_{\text{cl}}}$ and a rooted spanning tree. Enumerate vertices in a traversal that places every parent before its children. Store their labels, parent indices and tree-bond labels. Lemma C.2 leaves exactly $\beta(G) \leq K_{\text{cl}}$ residual edges; store each with increasing endpoint indices and sort these triples. Pad all unused coordinates with null states. This state satisfies Definition C.1, and the vertex enumeration is a label-preserving isomorphism between its decoded graph and G . $\square$

This result concerns the unrestricted representation. For program-specific alphabets $\mathcal{X}_i(P)$ and fixed values $x_{P,i}$, define

$$\Omega_P = \{x \in \bar{\mathfrak{X}}_{N,K_{\text{cl}}} : x_i \in \mathcal{X}_i(P) \forall i, \quad x_i = x_{P,i} \forall i \in \mathcal{F}(P)\}. \quad (33)$$

Its nonempty well-formed intersection defines $\mathcal{P}_{\text{adm}} = \{P : \bar{\mathfrak{X}}_{N,K_{\text{cl}}} \cap \Omega_P \neq \emptyset\}$. Nonemptiness alone establishes neither chemical validity nor satisfaction of every morphology constraint. Corpus admission additionally checks the annotated encoding, program alignment and exact decode round trip. Representational completeness makes no claim that every graph is encodable under every P .

C.2 EXACT ASSEMBLY AND CATALOGUE REACHABILITY

Definition A.1 distinguishes the complete input preimage $\mathcal{E}_c^*(G)$ from recovered input-trace pairs $\widehat{\mathcal{E}}_c(G)$. Fix a registered chemistry c and role-specific catalogues $\mathcal{C} = (\mathcal{C}_r)_r$. Let $\mathfrak{C}_c(\mathcal{C}) \subseteq \mathfrak{C}_c$ contain the admissible precursor inputs whose every role-assigned component belongs to its catalogue. The reachable product set is

$$\mathcal{G}_{\text{cat}}(c, \mathcal{C}) = \bigcup_{\mathbf{C} \in \mathfrak{C}_c(\mathcal{C})} \mathcal{F}_c(\mathbf{C}). \quad (34)$$

This definition includes all registered outcomes of the admissible inputs, not only products found by a bounded search.

Proposition C.4 (Finite-component reachability). *If $\mathfrak{C}_c(\mathcal{C})$ and each $\mathcal{F}_c(\mathbf{C})$ are finite, then*

$$|\mathcal{G}_{\text{cat}}(c, \mathcal{C})| \leq \sum_{\mathbf{C} \in \mathfrak{C}_c(\mathcal{C})} |\mathcal{F}_c(\mathbf{C})| \leq m_{\text{max},c}(\mathcal{C}) |\mathfrak{C}_c(\mathcal{C})|, \quad (35)$$

where $m_{\text{max},c}(\mathcal{C})$ is the largest outcome count, or zero for an empty input set. For every $G \in \mathfrak{G}_{N,K_{\text{cl}}}$,

$$G \in \mathcal{G}_{\text{cat}}(c, \mathcal{C}) \iff \mathcal{E}_c^*(G) \cap \mathfrak{C}_c(\mathcal{C}) \neq \emptyset. \quad (36)$$

Proof. The cardinality of a finite union is at most the sum of its member-set cardinalities; bounding each summand by its maximum proves both inequalities. If there are no inputs, all three quantities are zero. Membership in the union means that some catalogue input has G as a forward product, which is exactly the stated intersection condition. $\square$

Definition C.5 (Strong catalogue escape). An exactly assemblable graph strongly escapes $\mathcal{C}$ under c when every exact input uses at least one out-of-catalogue component. Its indicator is

$$\text{Escape}^*(G; c, \mathcal{C}) = \mathbf{1}[\mathcal{E}_c^*(G) \neq \emptyset, \quad \mathcal{E}_c^*(G) \cap \mathfrak{C}_c(\mathcal{C}) = \emptyset]. \quad (37)$$

A catalogue assembler whose outputs lie in $\mathcal{G}_{\text{cat}}(c, \mathcal{C})$ therefore has zero strong-escape yield. For a verified exact-L1 product, strong escape is equivalent to $G \notin \mathcal{G}_{\text{cat}}(c, \mathcal{C})$. This can be checked against a complete canonical product set or by exhaustive exclusion of all catalogue-only decompositions. A bounded reverse search cannot establish that exclusion from one novel witness.

The reported experiments did not construct a complete common-reference reachable product set. Their component-novelty metric concerns the selected verified input. Whole-graph generation removes catalogue membership from the generative representation; that architectural property alone proves neither strong escape nor positive probability of any particular out-of-catalogue graph.

C.3 CONDITIONAL FLOW AND IDEAL POPULATION RECOVERY

We prove Theorem 3.1 for the reference objective in Equation 7. Let $X_\star = \text{Enc}_P(G, \alpha)$ denote a clean training endpoint, and distinguish it from the time-indexed process X_t . The training-path measure is

$$\nu(dG, d\alpha, dP, dt, dx) = \mu_{\text{bal}}(dG, d\alpha, dP) \mathcal{U}_{(0,1)}(dt) q_{t|1}^P(dx \mid \text{Enc}_P(G, \alpha)). \quad (38)$$

For fixed P , the encoded endpoint law q_1^P induces

$$q_t^P(x) = \sum_{x_1} q_1^P(x_1) q_{t|1}^P(x \mid x_1), \quad \rho_i(a \mid x, t, P) = \text{Pr}_\nu[X_{\star,i} = a \mid X_t = x, t, P]. \quad (39)$$

Full source support on the finite allowed alphabets makes $q_t^P(x) > 0$ for every $x \in \Omega_P$ and $t < 1$. At $t = 0$ the path is q_0^P ; at $t = 1$ it is q_1^P .

Conditional and marginal rates. For $i \in \mathcal{V}(P)$ and $a, u, v \in \mathcal{X}_i(P)$, write $s_{t,i}^P(u | a) = (1-t)q_{0,i}^P(u) + t\mathbf{1}\{u=a\}$ and $\dot{s}_i^P(u | a) = \mathbf{1}\{u=a\} - q_{0,i}^P(u)$. The conditional rate used in the ideal analysis is the R^* construction of Qin et al. (2025), restricted to the allowed alphabet:

$$R_{t,i}^*(u, v | a, P) = \frac{[\dot{s}_i^P(v | a) - \dot{s}_i^P(u | a)]_+}{|\mathcal{X}_i(P)| s_{t,i}^P(u | a)}, \quad u \neq v, \quad 0 \leq t < 1. \quad (40)$$

Diagonal entries are the negative row sums. Denominators are positive for $t < 1$; no value at $t = 1$ is needed. Fixed coordinates have zero rates. For $x \neq y$ in Ω_P , the joint conditional and predicted rates are

$$R_t^*(x, y | x_1, P) = \sum_{i \in \mathcal{V}(P)} \mathbf{1}\{y_{-i} = x_{-i}\} R_{t,i}^*(x_i, y_i | x_{1,i}, P), \quad (41)$$

$$R_t^\theta(x, y | P) = \sum_{i \in \mathcal{V}(P)} \mathbf{1}\{y_{-i} = x_{-i}\} R_{t,i}^\theta(x_i, y_i | x, P). \quad (42)$$

Here x_{-i} omits coordinate i , and Equation 6 defines the predicted coordinate rate. Diagonal joint rates again make each row sum to zero. States differing at two or more coordinates have zero instantaneous transition rate.

Proposition C.6 (Program-fixed invariance). *Fix $P \in \mathcal{P}_{\text{adm}}$ and an endpoint in Ω_P . Corruption by Equation 5 stays in Ω_P . The ideal CTMC started from q_0^P stays there for $t < 1$ whenever its coordinate predictions are probability distributions on the allowed alphabets. Numerical sampling in Equation 9, started from q_0^P and using the allowed coordinate alphabets, satisfies $\Pr[Z_k \in \Omega_P \text{ for all } 0 \leq k \leq L_{\text{samp}}] = 1$.*

Proof. The corruption law has zero mass outside Ω_P , and the ideal generator has no transition to its complement. Each numerical coordinate draw stays in its allowed alphabet and Π_P restores the prescribed fixed values. Induction on k proves the discrete statement. These arguments do not require an accurate denoiser and do not imply molecular validity. $\square$

Proposition C.7 (Posterior minimizer of the reference loss). *Under the reference training measure ν , assume unrestricted measurable coordinate predictors and positive weights $w_i(P)$ for each variable coordinate. Every population minimizer of Equation 7 satisfies $d_{\theta,i}(a | x, t, P) = \rho_i(a | x, t, P)$ for all $a \in \mathcal{X}_i(P)$, ν -almost surely on contexts with $i \in \mathcal{V}(P)$.*

Proof. Condition on (X_t, t, P) and a variable coordinate. Since $w_i(P)$ is positive and does not depend on its clean target after this conditioning, its conditional loss is

$$w_i(P) [H(\rho_i) + D_{\text{KL}}(\rho_i \| d_{\theta,i})]. \quad (43)$$

The entropy is independent of the predictor, and the divergence is nonnegative with equality only at $d_{\theta,i} = \rho_i$. An unrestricted class can attain all these pointwise minima together. A population minimizer must attain them almost surely, since a positive-measure excess would increase the integrated loss. Fixed coordinates contribute no term. $\square$

Corollary C.8 (Population recovery of the local rate). *For the reference population minimizer in Proposition C.7, $R_t^{\theta^*}(x, y | P)$ equals the conditional expectation of $R_t^*(x, y | X_*, P)$ given $(X_t = x, t, P)$, almost surely under the path measure.*

Proof. Each local conditional rate depends on the endpoint only through its coordinate $X_{*,i}$. Substituting the posterior marginal into Equation 6 therefore computes the required conditional expectation. Summing over coordinates gives the joint rate. $\square$

Proof of Theorem 3.1. We verify the forward equation first for one coordinate, then for the conditional product path, and finally after averaging over endpoints. Fix P , x_1 and i ; suppress these indices in s_t and $\dot{s}$. For each v and $t < 1$, the net flux of Equation 40 is

$$\sum_u s_t(u) R_t^*(u, v) = \frac{1}{|\mathcal{X}_i(P)|} \sum_u ([\dot{s}(v) - \dot{s}(u)]_+ - [\dot{s}(u) - \dot{s}(v)]_+) \quad (44)$$

$$= \dot{s}(v) - \frac{1}{|\mathcal{X}_i(P)|} \sum_u \dot{s}(u) = \dot{s}(v). \quad (45)$$

The last equality uses $\sum_u \dot{s}(u) = 0$. The product rule then gives the conditional joint forward equation because the generator changes one coordinate at a time. Averaging over q_1^P and using Bayes' rule yields

$$\partial_t q_t^P(y) = \sum_{x, x_1} q_1^P(x_1) q_{t|1}^P(x | x_1) R_t^*(x, y | x_1, P) \quad (46)$$

$$= \sum_x q_t^P(x) \mathbb{E}[R_t^*(x, y | X_*, P) | X_t = x, t, P]. \quad (47)$$

Corollary C.8 identifies the last conditional expectation with the learned ideal rate for almost every t . On each interval $[0, 1 - \eta]$, $0 < \eta < 1$, source positivity bounds all denominators away from zero. The rates are bounded measurable functions on a finite state space, so the forward equation has a unique solution for initial law q_0^P . Since q_t^P is positive throughout Ω_P , almost-sure equality under the path measure implies rate equality at every state for Lebesgue-almost every t . Changes on the exceptional times do not change the integral forward equation. The exactly simulated process therefore has law q_t^P .

The finite product path converges to the endpoint point mass as $t \uparrow 1$. Summing over its finite endpoint support gives $q_t^P \rightarrow q_1^P$ in total variation. This establishes a limit of marginal laws; a pathwise value at the singular endpoint is not required. Exact decode round trip maps each endpoint to its source constitution, proving $\text{Dec}_\# q_1^P = \mu_G^P$. The assumed endpoint validity and verified L1 give the probability-one conclusion. If there are no variable coordinates, the compatible state space is a singleton and the same conclusions hold for the constant process. $\square$

Corollary C.9 (Decoder contraction and failure probability). *Let q_1^P satisfy the endpoint assumptions of Theorem 3.1. Let $\tilde{q}^P$ be any law on $\bar{\mathcal{X}}_{N, K_{\text{el}}} \cup \{\perp_{\text{term}}\}$, with q_1^P extended by zero mass at $\perp_{\text{term}}$ and $\text{Dec}(\perp_{\text{term}}) = \perp_{\text{dec}}$. Define $\mathcal{B}_P$ as states whose decode fails, is molecularly invalid or fails verified L1. Then*

$$\|\text{Dec}_\# \tilde{q}^P - \mu_G^P\|_{\text{TV}} \leq \|\tilde{q}^P - q_1^P\|_{\text{TV}}, \quad \tilde{q}^P(\mathcal{B}_P) \leq \|\tilde{q}^P - q_1^P\|_{\text{TV}}, \quad (48)$$

where $\|p - q\|_{\text{TV}} = \frac{1}{2} \sum_x |p(x) - q(x)|$.

Proof. Every decoder-output event has a preimage in the raw space, so its probability difference is bounded by the raw-state total variation. Taking the supremum over output events proves the first bound. The endpoint assumptions give $q_1^P(\mathcal{B}_P) = 0$; applying the same event bound to $\mathcal{B}_P$ proves the second. $\square$

The corollary can be applied to the law *after* numerical sampling and terminal readout. It does not estimate that law's distance from q_1^P . In particular, error before the learned terminal readout cannot be substituted without controlling the readout's effect.

C.4 ACCEPTED COUNTS UNDER A FIXED SAMPLING BUDGET

Fix a trained model, requested chemistry c and its program-sampling policy. An attempt outcome Y contains its sampled program P and decoded graph or typed failure. Let Q be the common law of these outcomes and define

$$A(Y) = \mathbf{1}\{G \text{ is valid and } \widehat{\text{L1}}(G, c) = 1\}, \quad z_Q = \mathbb{E}_Q[A(Y)], \quad (49)$$

with $A(Y) = 0$ for failures. This definition permits a fresh morphology-specific P on every attempt. A comparison with a single fixed P is the corresponding special case.

Proposition C.10 (Exact law of matched-budget post-hoc filtering). *For an integer $B \geq 0$ and independent attempts with common law Q , the accepted-attempt count $N_B = \sum_{b=1}^B A(Y_b)$ is binomial with parameters (B, z_Q) . For all $z_Q \in [0, 1]$,*

$$\mathbb{E}[N_B] = Bz_Q, \quad \text{Var}(N_B) = Bz_Q(1 - z_Q), \quad \Pr[N_B = 0] = (1 - z_Q)^B. \quad (50)$$

For $B = 0$ the last probability is one, including when $z_Q = 1$. If $z_Q > 0$, the conditional law of an accepted attempt is $Q_{\text{post}}(y) = Q(y)A(y)/z_Q$, and the number T_m of attempts needed for $m \geq 1$ acceptances satisfies $\mathbb{E}[T_m] = m/z_Q$. For two proposal laws under the same attempt budget, $\mathbb{E}[N_B^{(1)} - N_B^{(0)}] = B(z_{Q_1} - z_{Q_0})$.

Proof. Each acceptance indicator is Bernoulli with parameter z_Q , and independence makes their sum binomial. Its expectation and variance give the first two identities; all B attempts fail with probability $(1 - z_Q)^B$. For $z_Q > 0$, conditioning on acceptance gives the normalized law by Bayes’ rule. The waiting times between acceptances are independent geometric variables on $\{1, 2, \dots\}$ with mean $1/z_Q$; summing m of them proves the waiting-time formula. Subtracting the expected counts proves the comparison identity, without requiring independence between the two arms. $\square$

If $z_Q = 0$, accepted outcomes have no conditional law and $T_m = \infty$ almost surely for $m \geq 1$. The proposition counts accepted attempts, including duplicates. It does not establish that program conditioning increases acceptance; that is an empirical comparison. If requested programs are fixed in unequal strata, independent acceptance indicators generally have different probabilities and their sum is Poisson-binomial. Retraining variation is separate from both sampling laws.

C.5 ROUTE TERMINATION AND EVIDENCE COMPLETENESS

Lemma C.11 (Termination of bounded route assessment). *Fix a route contract $\mathfrak{B}$ with finite maximum depth $D_{\mathfrak{B}}$. Assume finitely many proposed expansions per molecule and reactants per expansion, a terminating evidence and forward-verification call for each proposal, and a search that assesses each proposed expansion finitely many times. For an admitted root trace, recursive assessment terminates with a dossier or typed failure in $\mathfrak{D} \cup \mathfrak{F}_{\mathfrak{B}}$.*

Proof. Induct on remaining depth. At depth zero, the terminating availability query either admits a terminal or returns a failure without further expansion. At positive remaining depth, only finitely many proposals are assessed, each using terminating local calls. Every proposed child has one fewer remaining level, so its assessment terminates by induction. A finite number of finite child assessments and proposals therefore terminates at the parent. The same argument applies if a finite outer root loop is used. Rejecting ancestor identities prevents circular dependencies from being admitted; the finite depth and no-infinite-retry assumptions establish termination. $\square$

Implementation invariant C.12 (Synthesis-dossier admission). *Under the frozen contract $\mathfrak{B}$, a successful assessor return $T \in \mathfrak{D}$ is admitted only after checking $\text{Complete}_{\mathfrak{B}}(G, T, P) = 1$. Every other outcome is recorded as a typed failure. This is the admission rule in Algorithm 1, not a theorem that bounded search finds every supported chemical route.*

The predicate checks molecular validity, a unique verified root trace, a well-formed finite tree, evidence admission and forward replay for every preparation step, and availability for every leaf. Its factors are defined in Appendix A.6. A failure establishes only that the assessor did not return a complete dossier under the recorded contract. It does not exclude a route under a larger budget, other admitted chemistry or another supplier snapshot.

D ASSEMBLY FAMILIES AND REACTION SCHEMES

The assembly schemes depict the chemistry introduced in Section 2.2. The 22-family structure atlas pairs products with precursor origins and exact assembly traces (Figure 4).

D.1 ASSEMBLY SCHEMES

The precursor roles and assembly patterns are summarized in Table 21.

Table 21: **Registered assembly families and precursor roles.** Source role names are retained; a family can contain multiple admitted program variants and role aliases. A list of role names does not specify that every role occurs in a single reaction. The atom-mapped program determines the participating components and transformation order.

Family	Registered source roles
A3 amine–aldehyde–alkyne	aldehyde, alkyne, amine head
Acid–epoxide diester	amine acid head, epoxide, hydrophobic acid
AEMA	AEMA, amine core, thiol periphery
Aldehyde Ugi 3-CR	aldehyde side, amine side, isocyanide side
Aldehyde Ugi 4-CR	aldehyde, amine, carboxylic acid, isocyanide
α -Isocyanoester dihydroimidazole	amine head, coupled ketone, isocyanide
Amine alkylation	amine head, bromoester arm
Amine–epoxide	amine head, epoxide tail
Aryl reductive amination	amine head, coupled aldehyde
Aza-Michael acrylamide	acceptor tail, amine head
Aza-Michael acrylate	acceptor tail, acrylate, alkyl acrylate, amine, amine head, amine nucleophile, amine precursor, conjugated carbonyl acceptor
Disulfide Michael	amine head, disulfide acceptor
Epoxide O-acylation	acyl tail, amine head, epoxide tail
iPhos	amine head, phosphate tail
Ketone–isocyanide amide	amine head, coupled ketone, isocyanide
Ketone Ugi 4-CR	amine, amine head, carboxylic acid, coupled ketone, isocyanide, ketone
Maleate	amine head, maleate
O-esterification	acid tail, aminoalcohol head
Passerini	aldehyde tail, amine acid head, isocyanide tail
Preassembled thiol-yne	alkynoic linker, amine head, thiol tail
Reductive amination	amine head, coupled aldehyde
Thiolactone	acrylate tail, amine head, thiolactone region

D.2 REACTION SCHEMES ACROSS THE ASSEMBLY FAMILIES

Figures 5–15 show the registered transformations for all 22 families. The companion schemes use exact-replay-verified representative assemblies. Unchanged hydrocarbon fragments are abbreviated by *R* labels local to each arrow; heteroatoms, ring systems and the connecting molecular framework are retained. Ordered stages use the preceding intermediate after the indicated repeated incorporations. Auxiliary reagents, salts, work-up and stereochemistry absent from the constitutional record are omitted. The drawings specify reaction-enumerated support without assigning experimental outcomes to the depicted combinations. Precursor-origin boundaries can differ from the functional headgroup, linker and tail regions introduced in Section 2.1. An L1 pass requires forward replay on the recovered precursors to reproduce the complete constitutional lipid graph exactly.

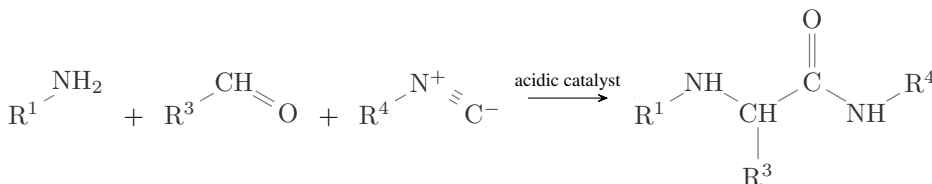
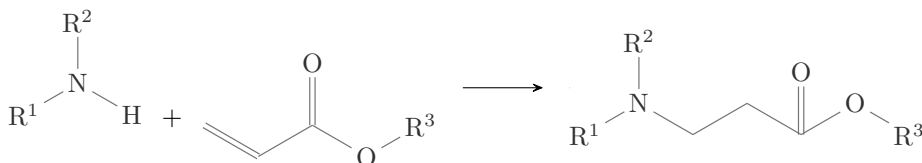
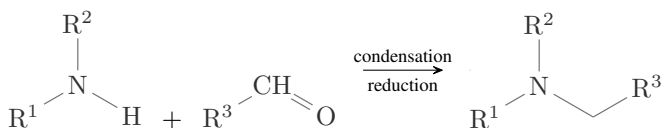
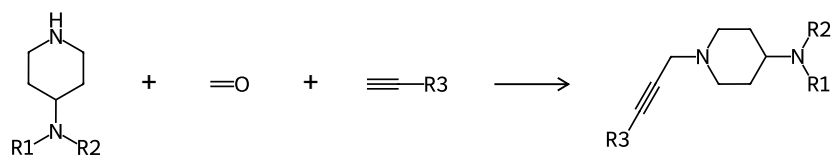
Ugi three-component reaction*amine head, aldehyde and isocyanide combine in one step; no carboxylic-acid reactant component***Repeated aza-Michael addition***one addition shown; repeat at registered N-H sites; the ester is unchanged***Repeated reductive amination***condensation and reduction form the carbon-nitrogen linkage*

Figure 5: **Representative assembly chemistries.** Monochrome skeletal drawings show the reactants and products of each registered assembly. Hydrogens are drawn only on the atoms whose substitution decides whether a product can be decomposed back into precursors: the head N-H, and the α -carbon C-H that the Ugi reaction takes from the aldehyde. R^i stand for the source-linked components of the frozen library, which are larger than the fragments drawn here. The Ugi variant has no carboxylic-acid reactant component.

a A3 amine–aldehyde–alkyne

A3 coupling

**b Aldehyde Ugi four-component assembly**

Ugi 4-CR

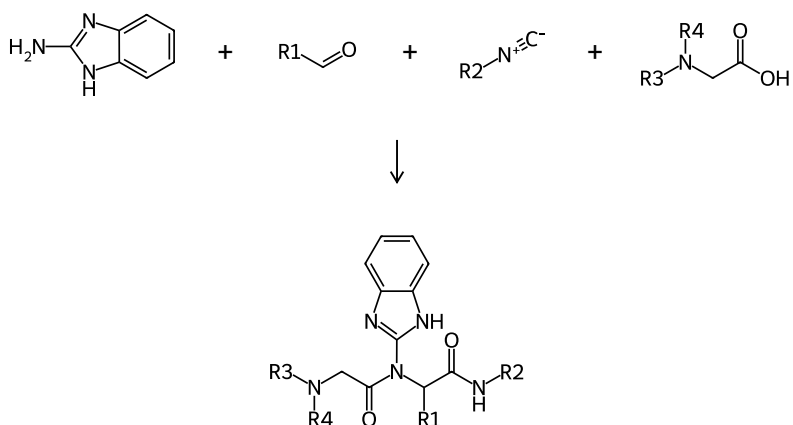


Figure 6: **Multicomponent assembly: A3 and Ugi four-component reactions.** **a**, A3 coupling creates C–N and C–C bonds while retaining the alkyne. **b**, Aldehyde Ugi 4-CR combines an amine, aldehyde, isocyanide and carboxylic acid. The acid supplies the N-acyl unit in the four-component reaction.

a Ketone Ugi four-component assembly

Ugi 4-CR

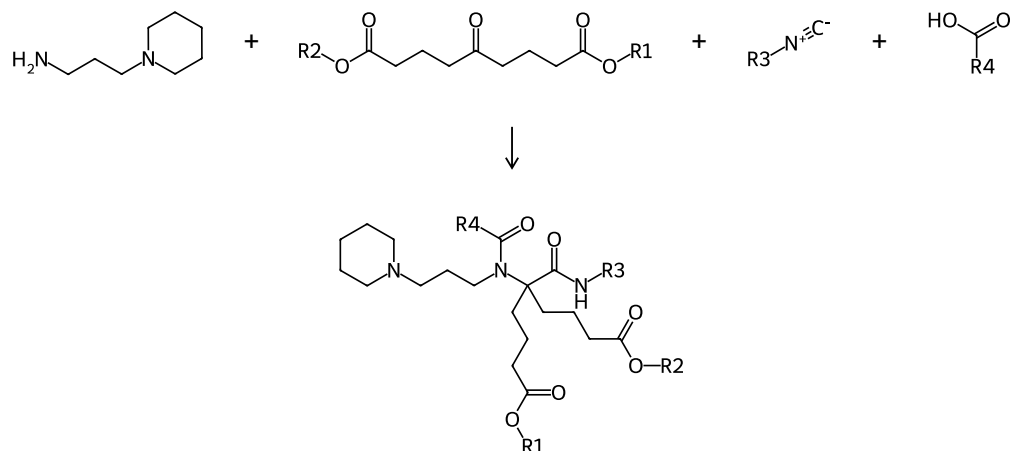
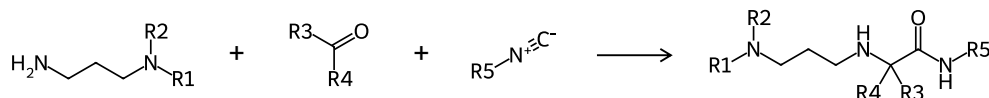


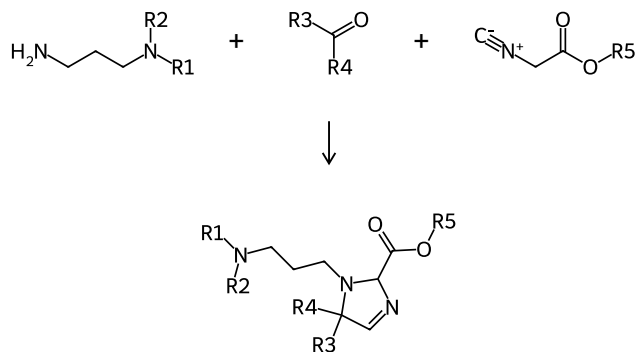
Figure 7: **Ketone Ugi four-component assembly.** The ketone reacts with an amine, isocyanide and carboxylic acid. Both carbon substituents remain attached to the ketone-derived center; the preassembled ester-bearing arms are retained, and the acid supplies the N-acyl unit.

a Ketone–isocyanide aminoamide

Three-component aminoamide assembly

**b α -Isocyanoester dihydroimidazole**

Condensation and cyclization

**c Passerini three-component assembly**

Passerini 3-CR

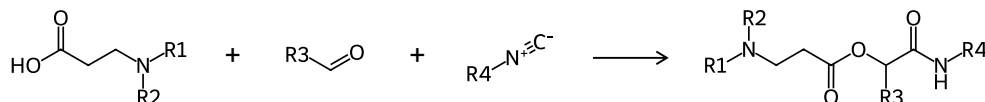
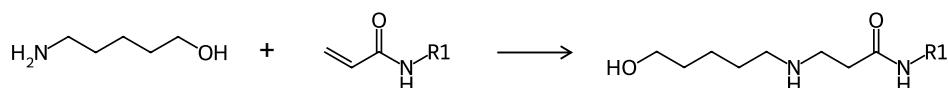


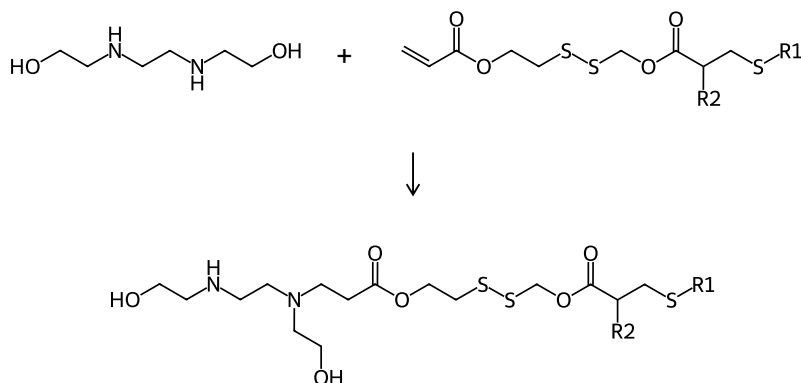
Figure 8: **Multicomponent assembly: aminoamide, dihydroimidazole and Passerini reactions.** **a**, The ketone–isocyanide branch forms an aminoamide from three components. **b**, An α -isocyanoester participates in dihydroimidazole ring formation, retaining its ester. **c**, Passerini assembly produces an α -acyloxy amide from an aldehyde, acid and isocyanide; the acid-derived acyl group attaches through oxygen.

a Aza-Michael addition to acrylamide

Aza-Michael addition; 2 incorporated events, first shown

**b Disulfide Michael addition**

Aza-Michael addition; S-S retained; 2 incorporated events, first shown

**c Maleate addition**

Aza-Michael addition (N-addition branch)

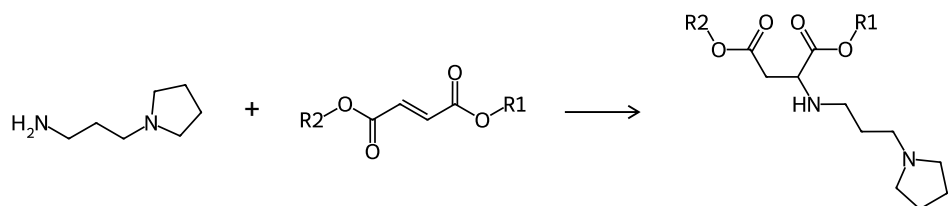
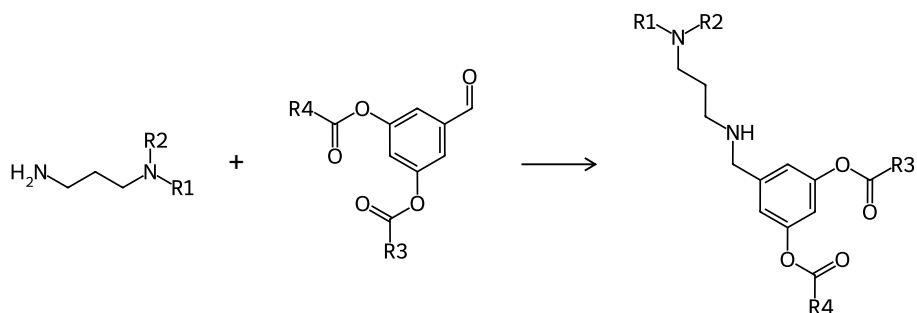


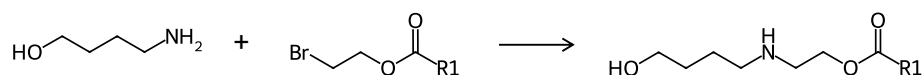
Figure 9: **Conjugate-addition families.** **a**, Amine addition to an acrylamide retains the amide interface. **b**, Addition to a disulfide-containing acrylate retains its S-S bond. **c**, The illustrated maleate program uses the N-addition branch and retains both ester groups. For repeated programs, one event is shown and the incorporated-event count is indicated. The aza-Michael acrylate reaction is shown in Figure 5.

a Aryl reductive amination

Condensation, then reduction

**b Amine alkylation**

N-alkylation; 2 incorporated events, first shown

**c Amine-epoxide opening**

Amine-epoxide ring opening; 2 incorporated events, first shown

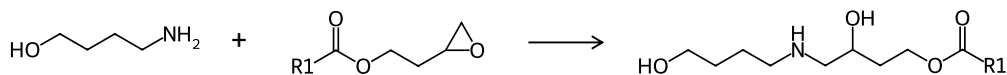
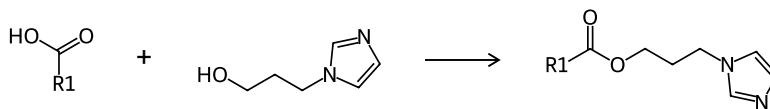


Figure 10: **Nitrogen functionalization.** **a**, Aryl reductive amination combines condensation with reduction. **b**, N-alkylation replaces the alkyl bromide leaving group. **c**, Amine attack opens an epoxide and retains the resulting alcohol. Repeated incorporations are labeled.

a O-Esterification

O-esterification

**b iPhos ring opening**

Cyclic-phosphate ring opening; deprotonated product

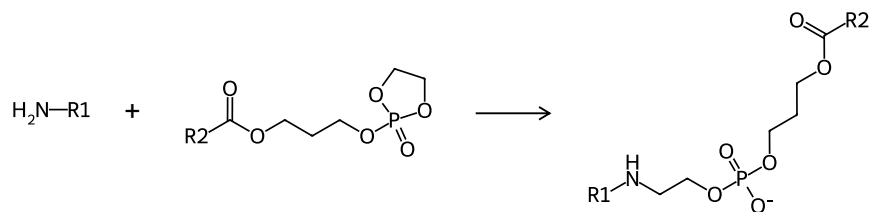
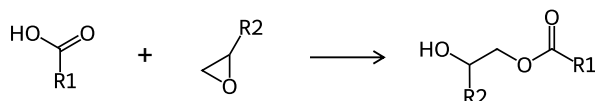


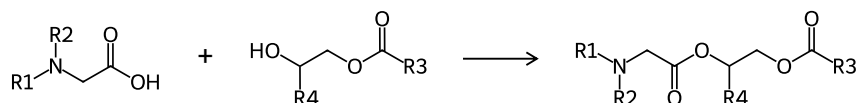
Figure 11: **Esterification and phosphate opening.** **a**, O-esterification joins an aminoalcohol to an acid while retaining the basic nitrogen. **b**, iPhos assembly opens a C–O bond of a cyclic phosphate to form the N–CH₂CH₂–O–P connection. Its deprotonated product representation is preserved.

a Acid-epoxide opening, then esterification

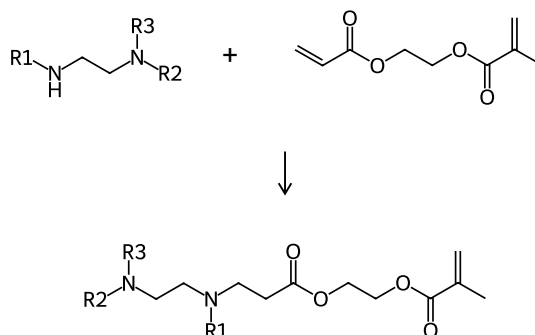
1. Acid-epoxide opening



2. O-esterification

**b AEMA: aza addition, then thiol addition**

1. Aza-Michael at acrylate



2. Thiol addition at methacrylate

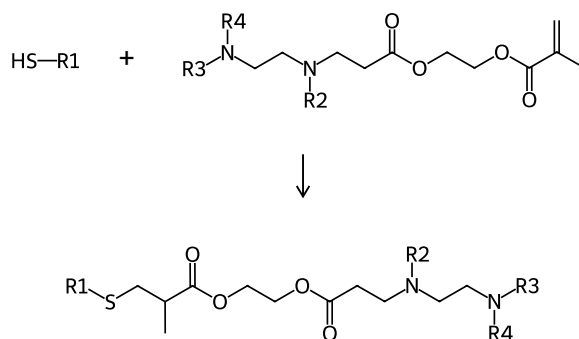
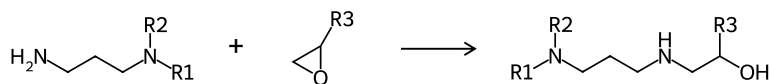


Figure 12: **Ordered assembly through hydroxy-ester and dual-acceptor intermediates.** **a**, Acid-epoxide opening first produces a hydroxy ester; subsequent esterification attaches the amino acid head through its carboxyl group. **b**, The AEMA program consumes the acrylate in an aza-Michael addition, then the remaining methacrylate in a thiol addition. Each second-stage reactant is the corresponding first-stage intermediate; *R* abbreviations are local to each arrow.

a Amine–epoxide opening, then O-acylation

1. Amine–epoxide opening; 2 incorporated events, first shown



2. O-acylation; 2 incorporated events, first shown

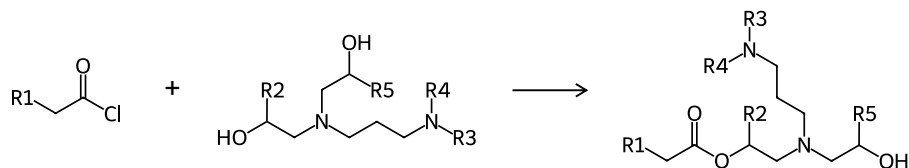


Figure 13: **Epoxide opening followed by O-acylation.** Amine–epoxide additions first form aminoalcohol arms. The resulting alcohols then react with acyl chlorides to form ester linkages. The representative program incorporates two epoxides and two acyl groups; one event from each stage is drawn, with its multiplicity indicated.

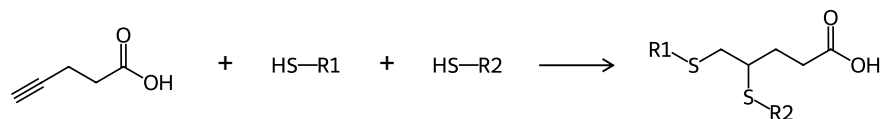
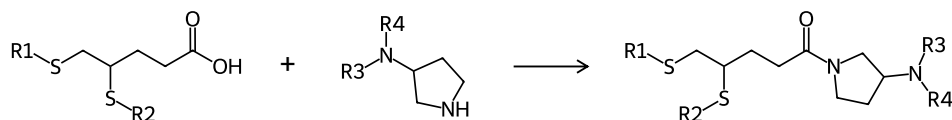
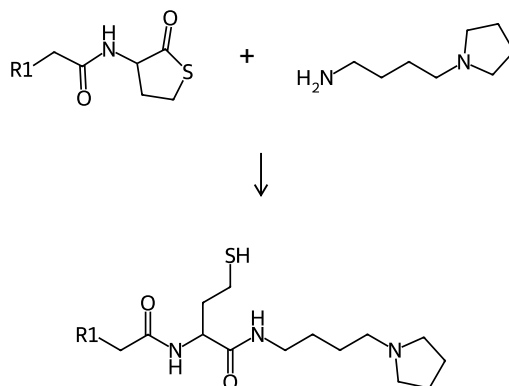
a Thiol-yne tail assembly, then amidation**1. Bis-thiol-yne addition****2. Amidation**

Figure 14: **Thiol-yne tail assembly followed by amidation.** Two identical thiols ($R_1 = R_2$ within each arrow) add across a terminal alkyne to form the bis(thioether) acid, which then undergoes amidation. The first arrow denotes the net double addition.

a Thiolactone aminolysis, then thiol-Michael addition

1. Thiolactone aminolysis



2. Thiol-Michael addition

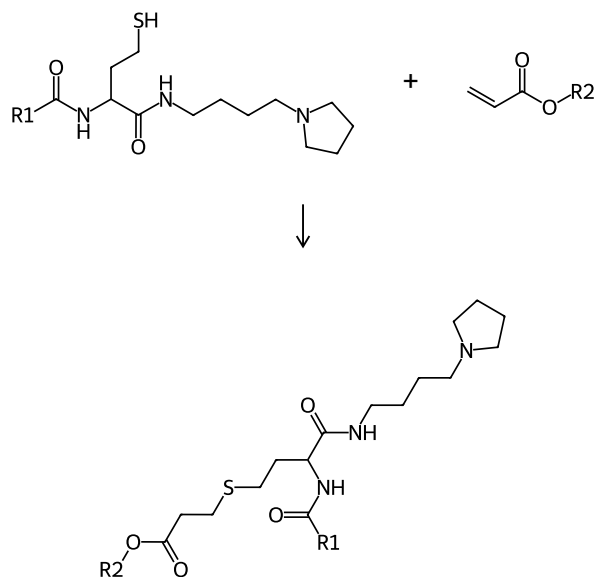


Figure 15: **Thiolactone aminolysis followed by thiol-Michael addition.** Aminolysis opens the thiolactone to form an amide and release a thiol. The subsequent addition to an acrylate forms a thioether. The retained sulfur framework distinguishes this transformation from disulfide formation.